On the precision of dynamic gene regulation

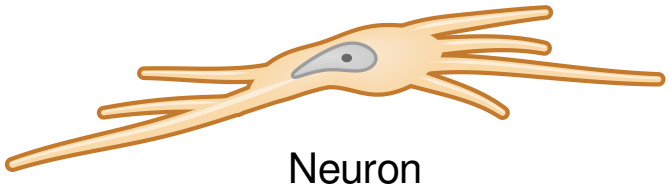
Tarek Tohme

Under the supervision of

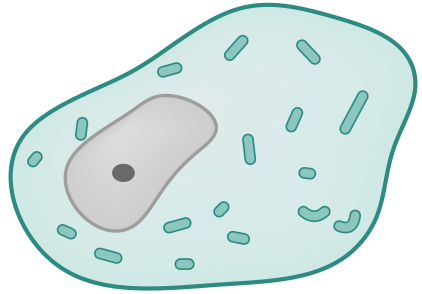
Aleksandra Walczak and Thierry Mora

November 4, 2026



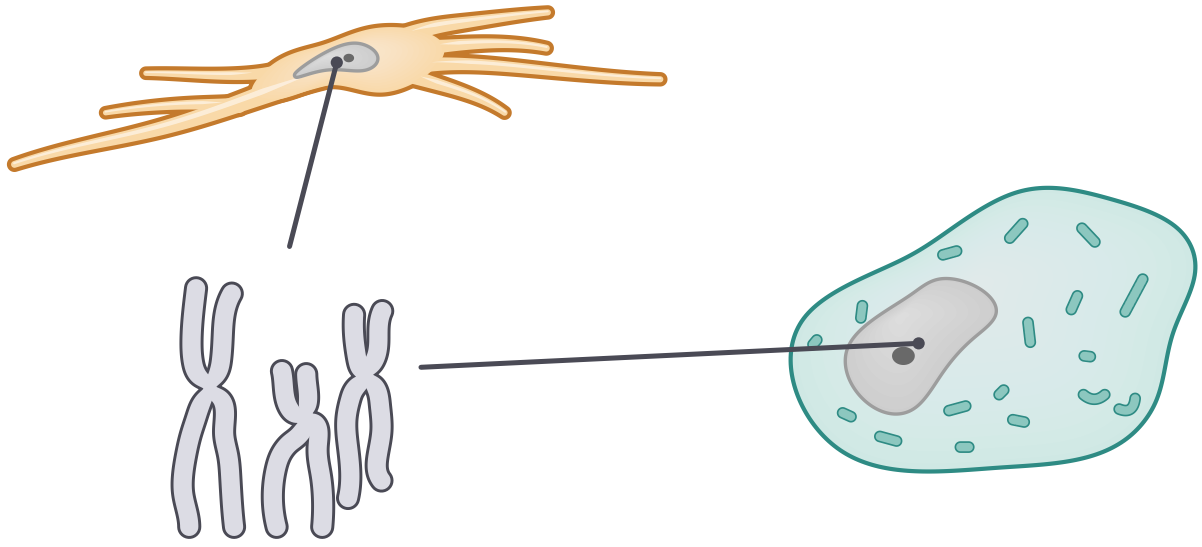


Neuron



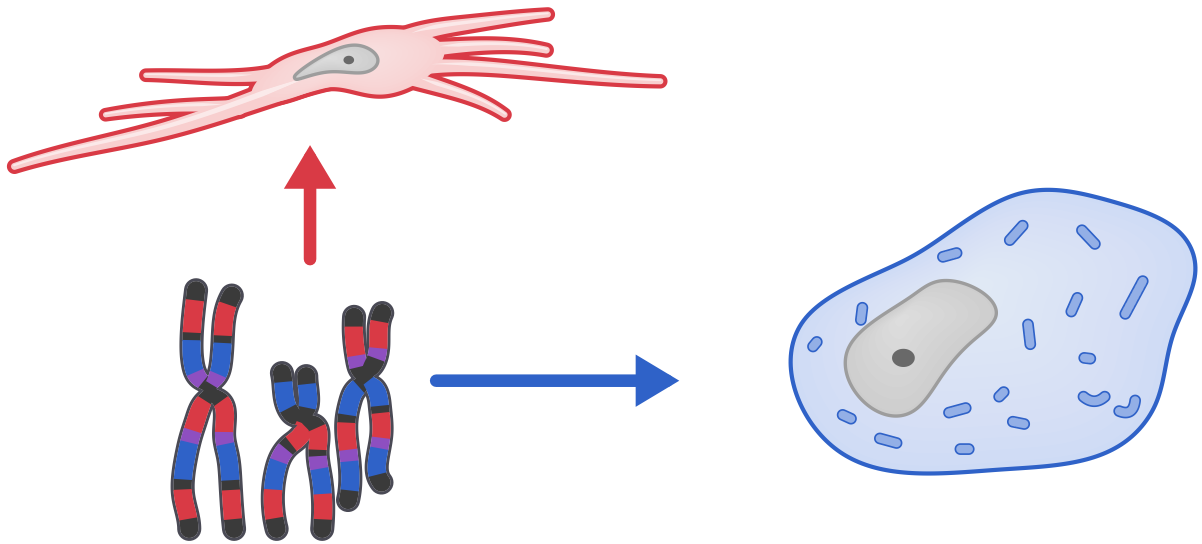
Kidney cell

scene1



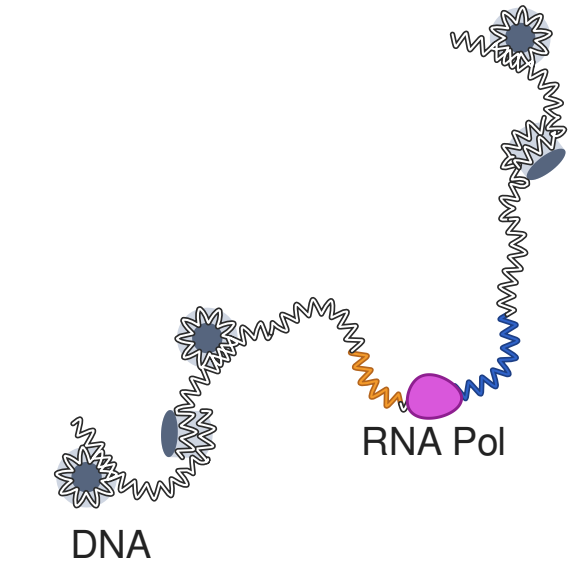
scene2

Gene expression is regulated



scene3

Overview of gene regulation

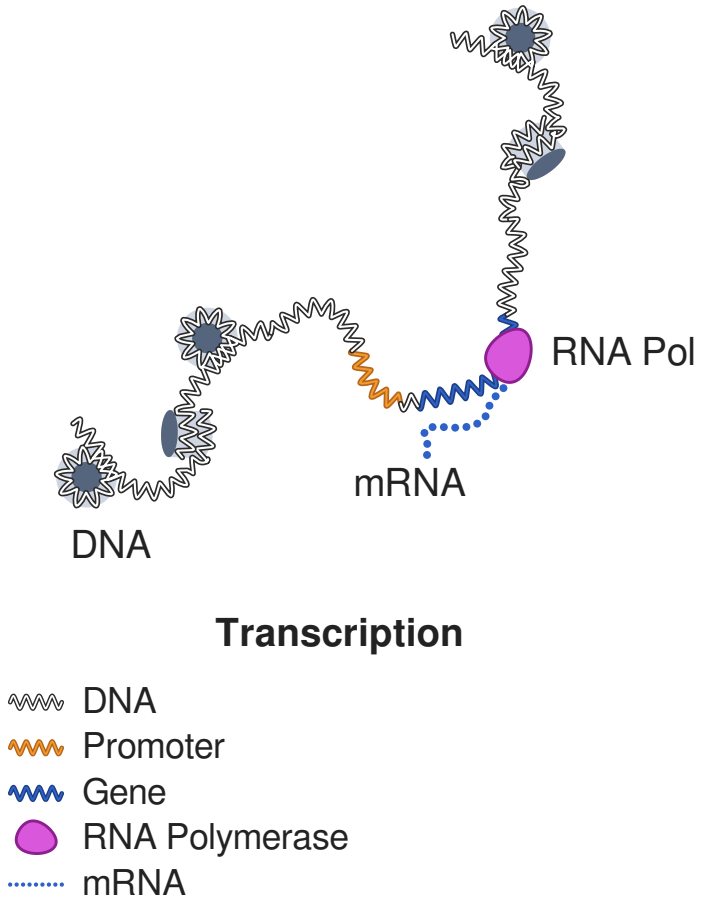


Transcription

DNA
Promoter
Gene
RNA Polymerase

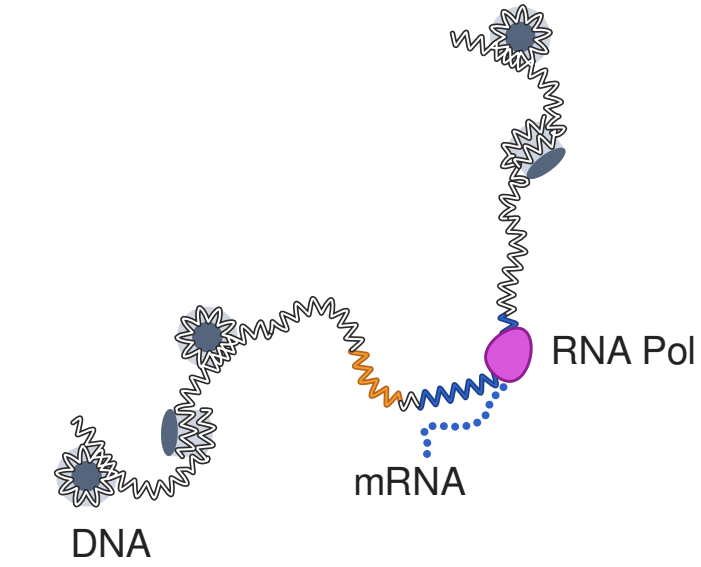
scene4_1

Overview of gene regulation

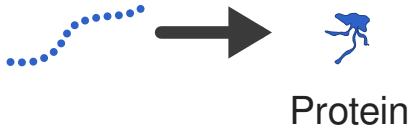


scene4_2

Overview of gene regulation



Transcription

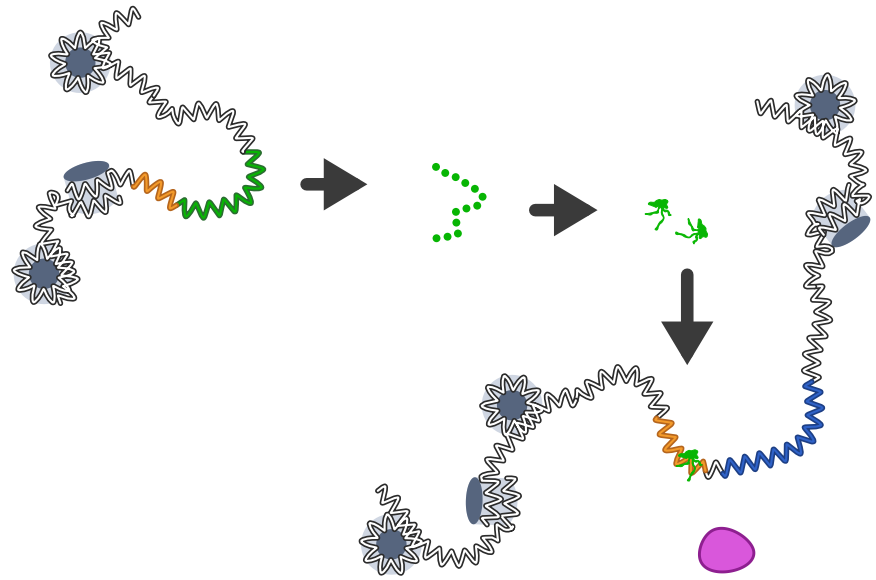


Translation

DNA
Promoter
Gene
RNA Polymerase
mRNA

scene4$_3$

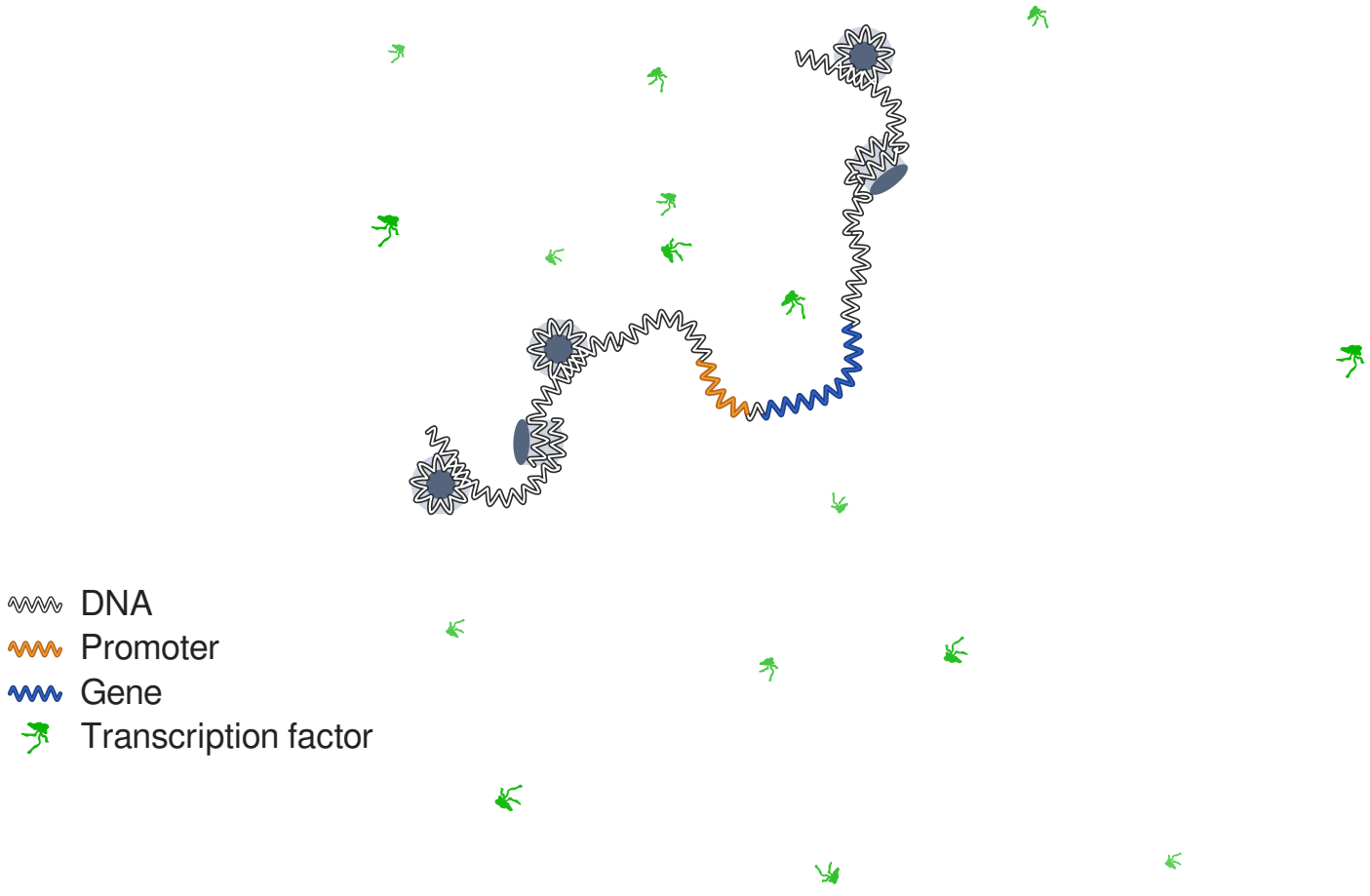
Overview of gene regulation



DNA
Promoter
Gene
Transcription factor
RNA Polymerase
mRNA

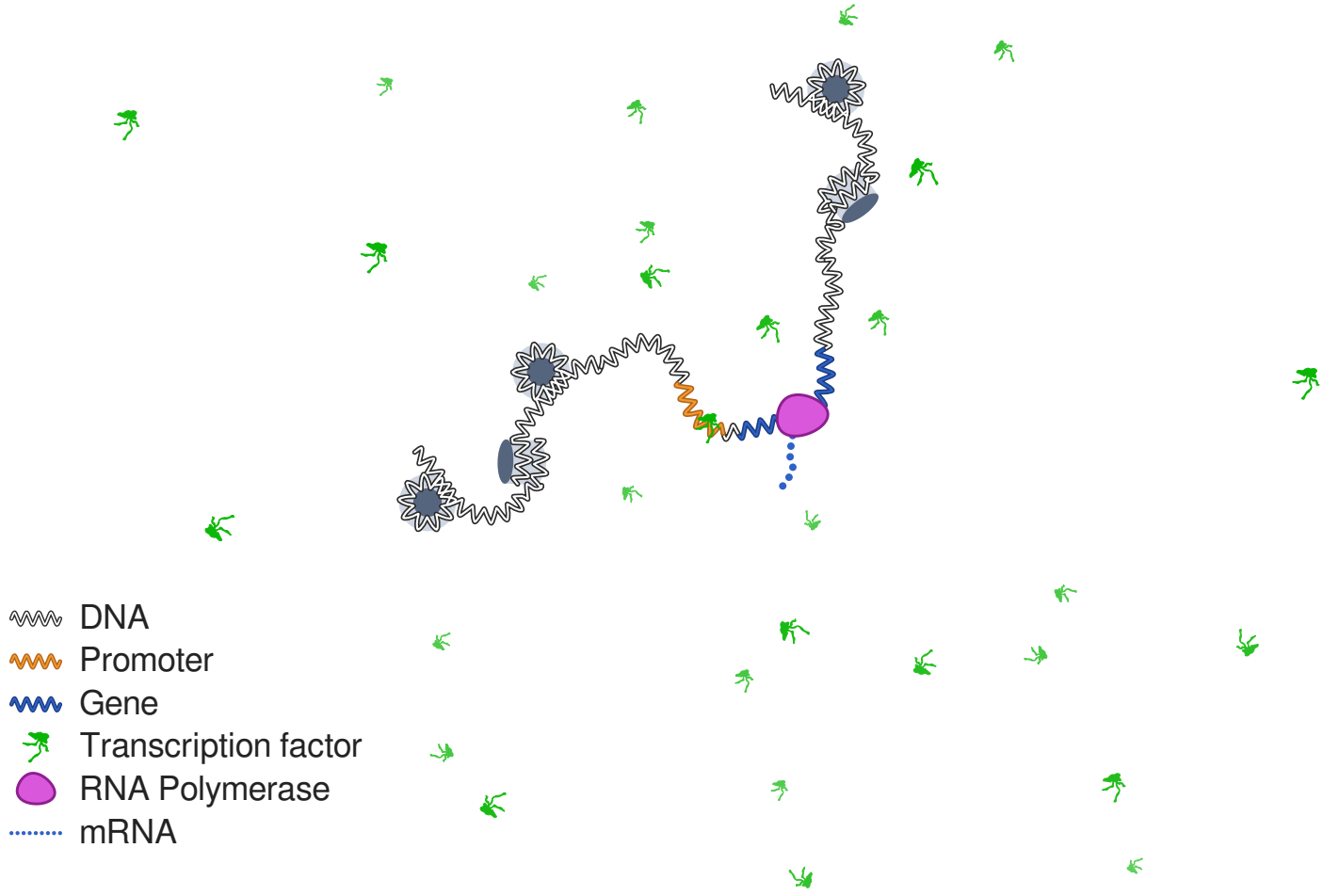
scene5

Expression levels are continuous ...



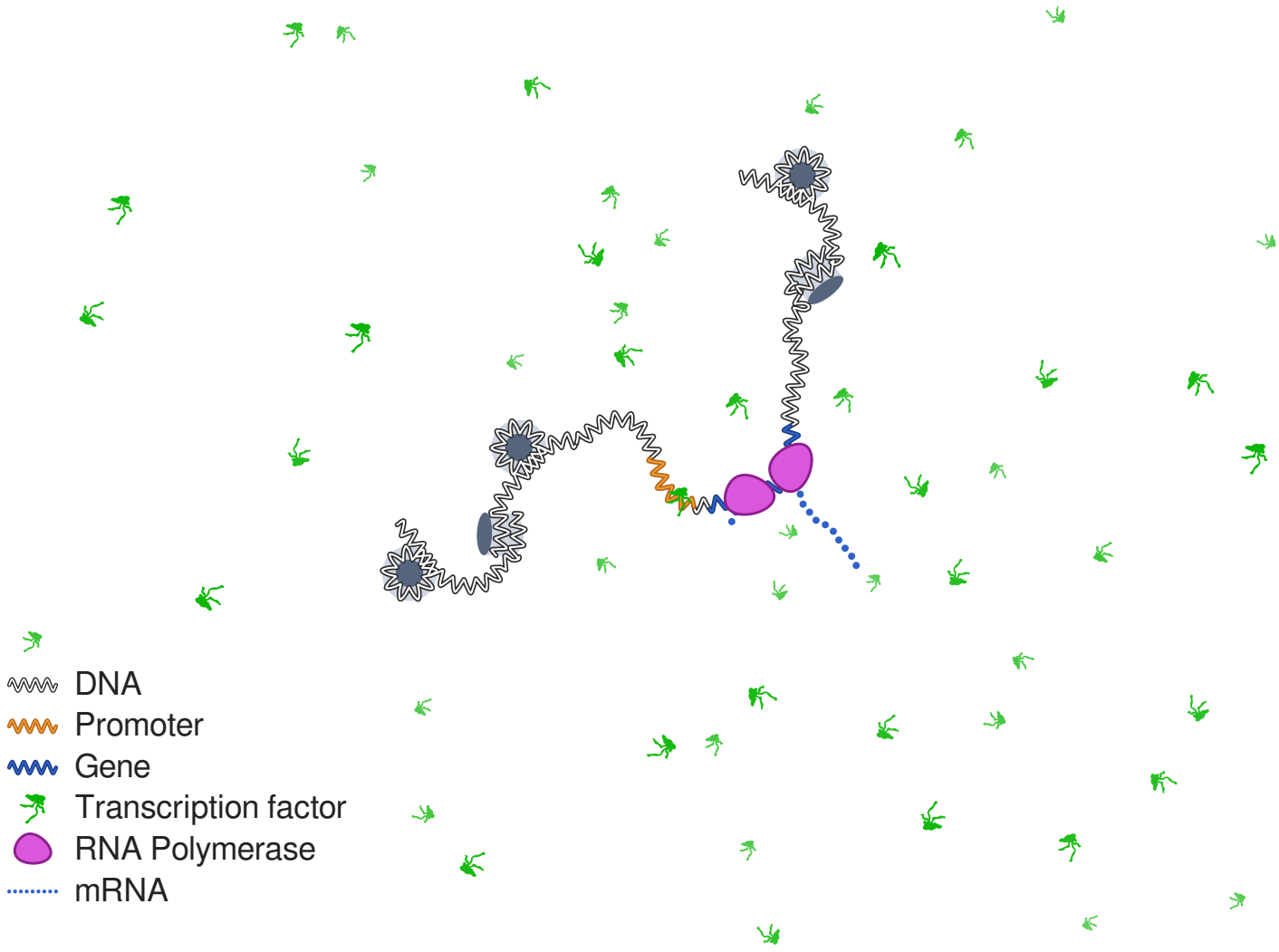
scene6$_1$

Expression levels are continuous ...



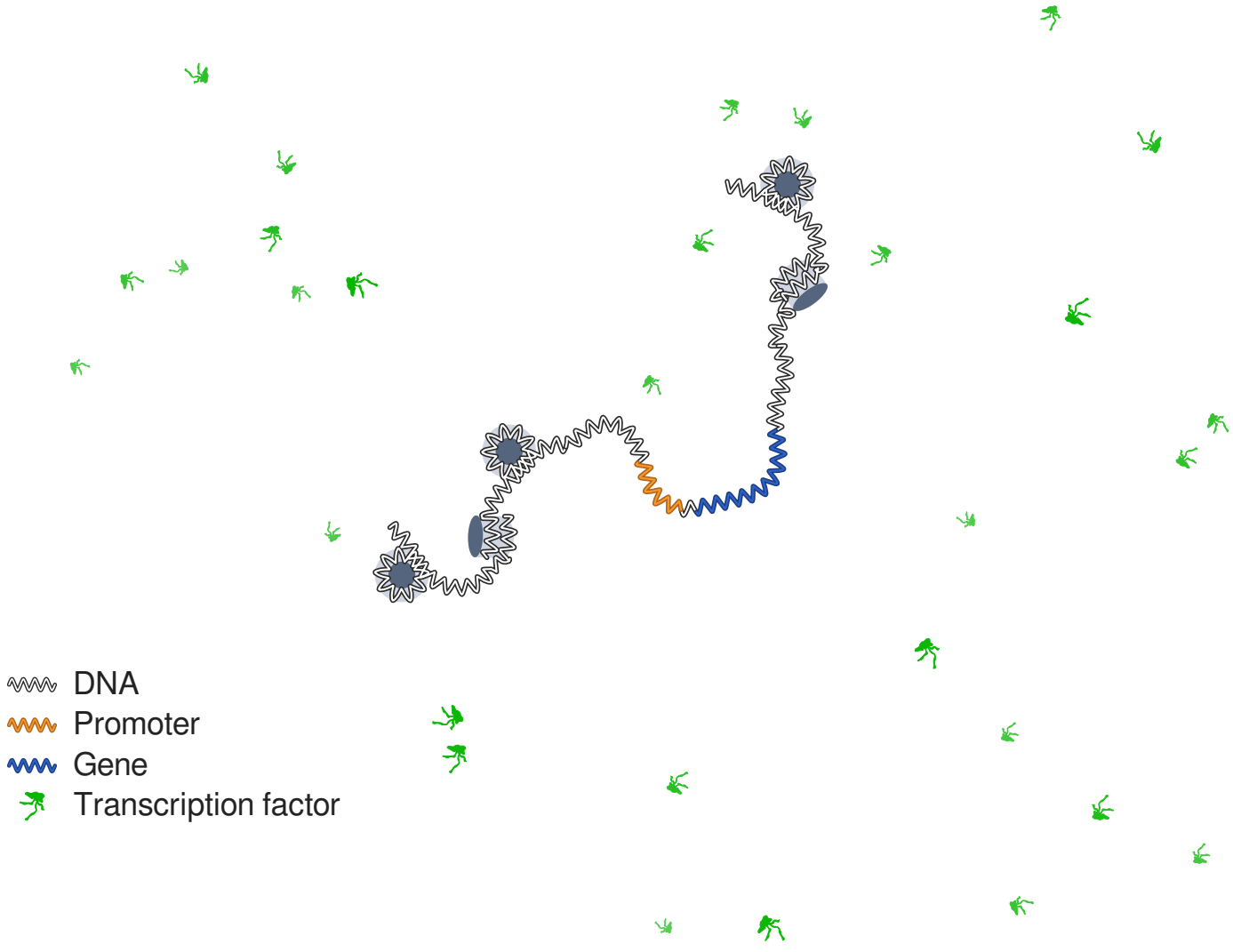
scene6_2

Expression levels are continuous ...



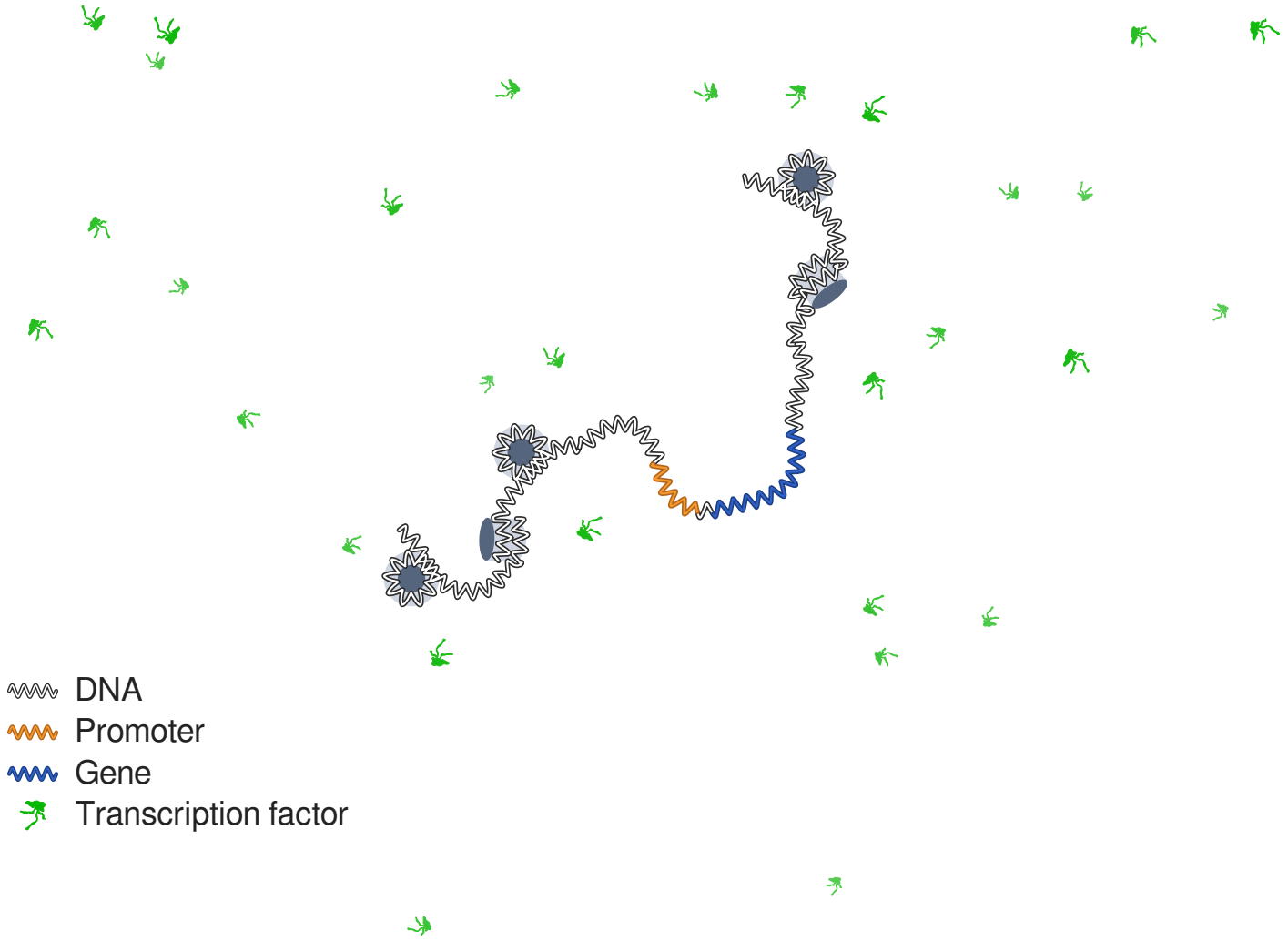
scene6$_3$

... but transcriptional control is noisy



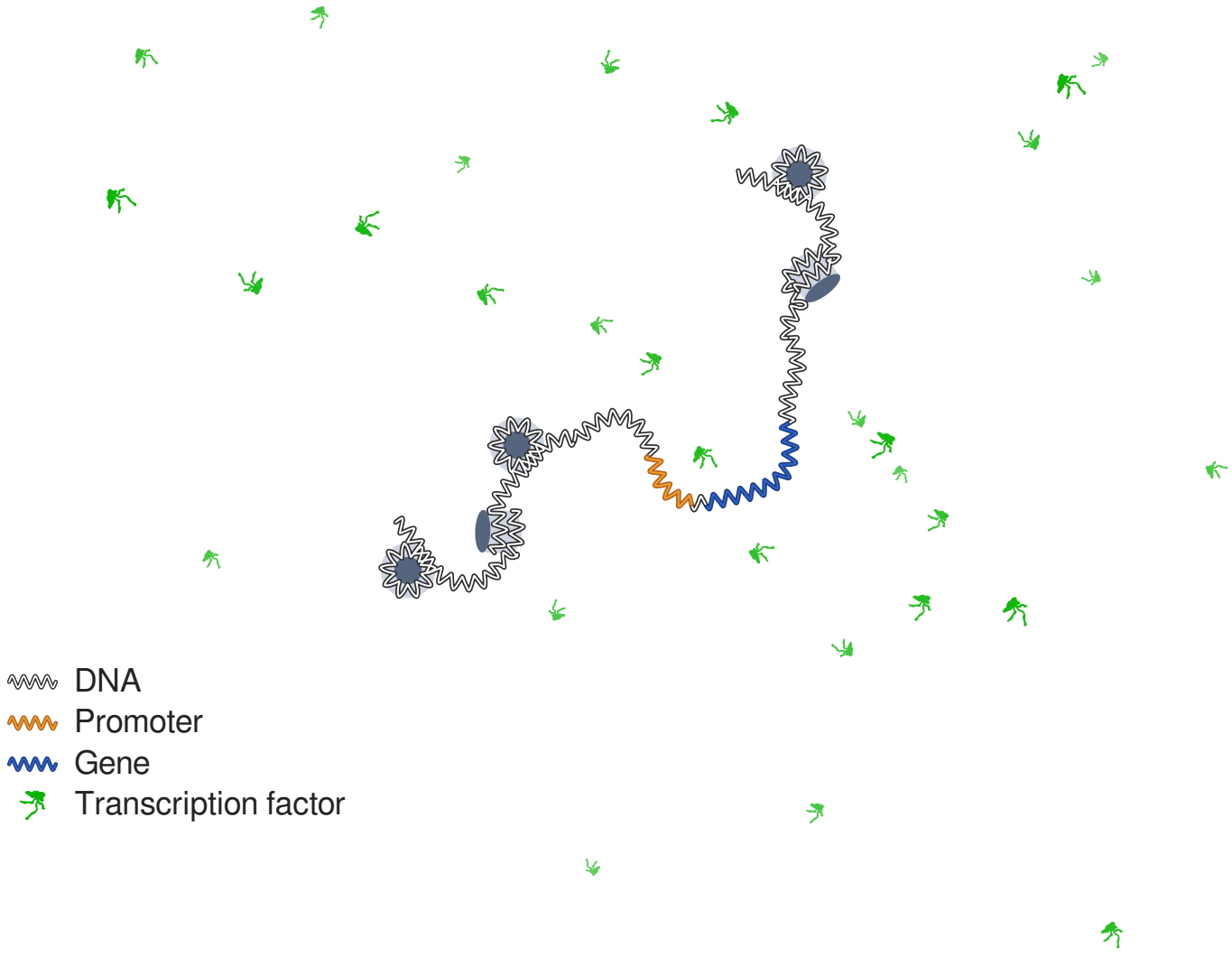
scene6$_4$

... but transcriptional control is noisy



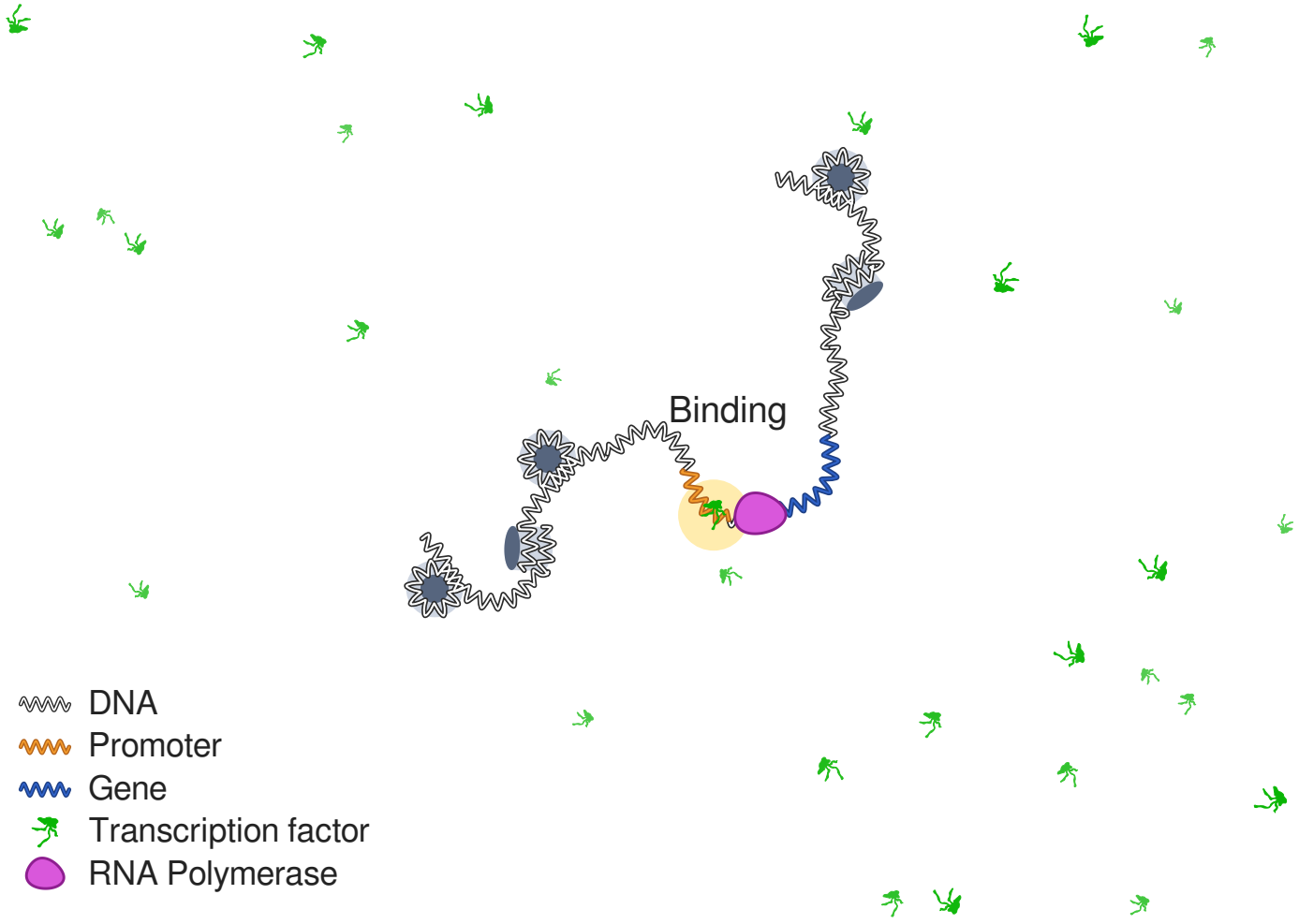
scene6_5

... but transcriptional control is noisy



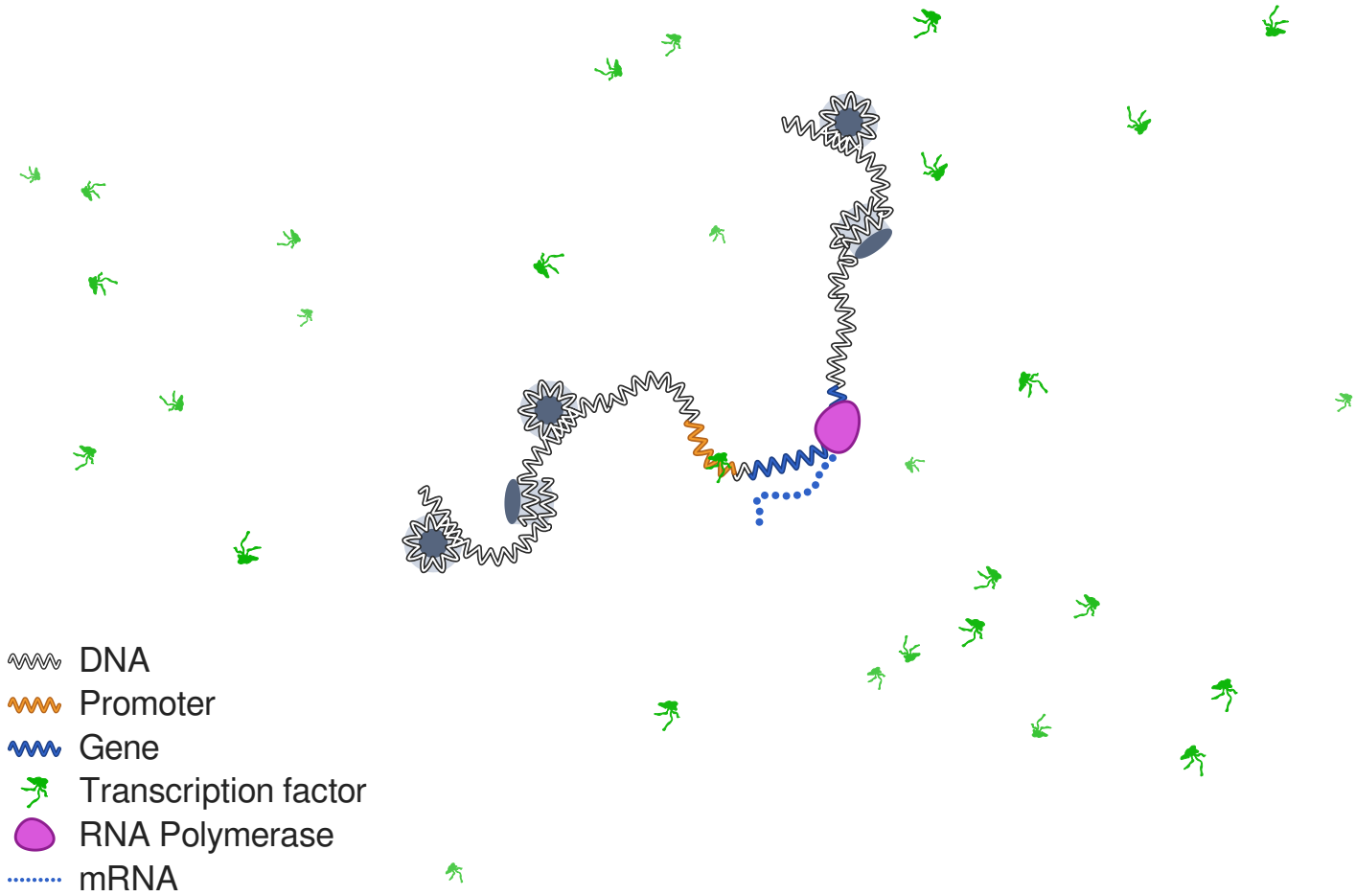
scene6_6

... but transcriptional control is noisy



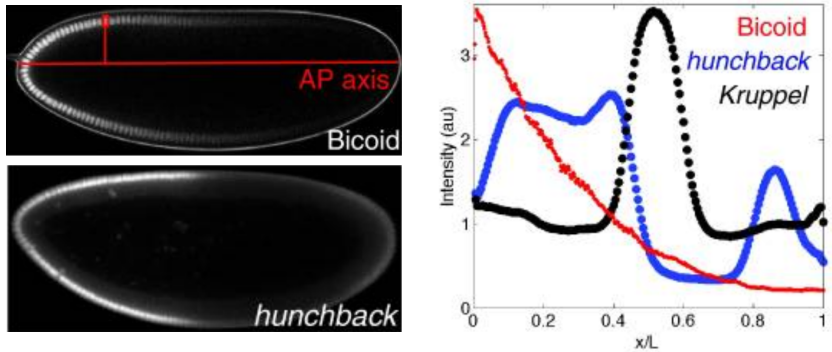
scene6$_7$

... but transcriptional control is noisy



scene6$_8$

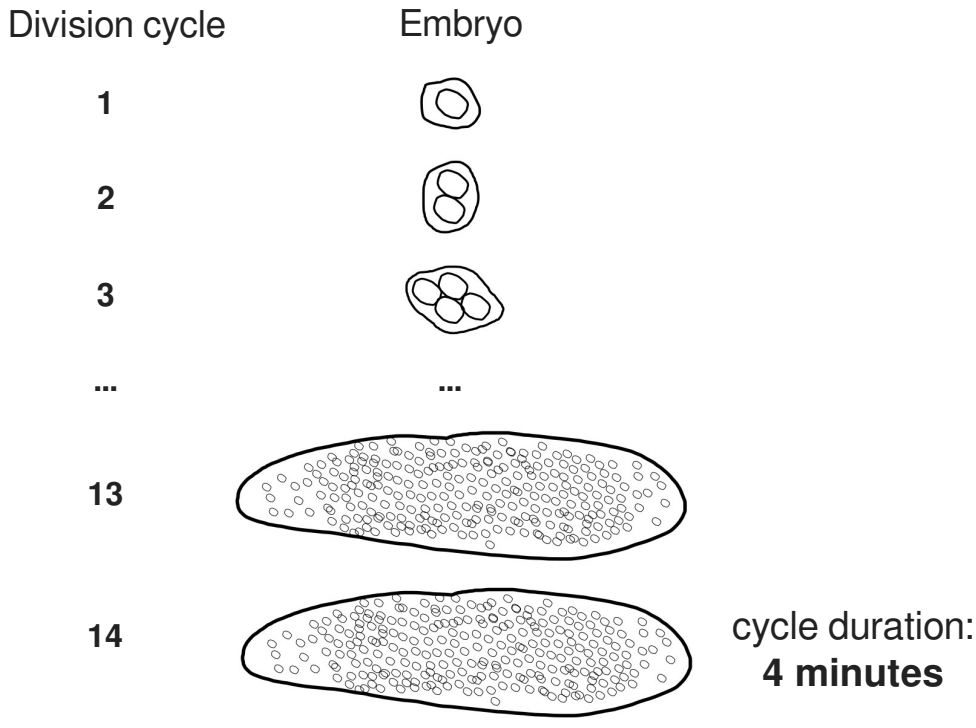
Sensitive response to shallow concentration gradients



Grimm et al., *Development,* 2010

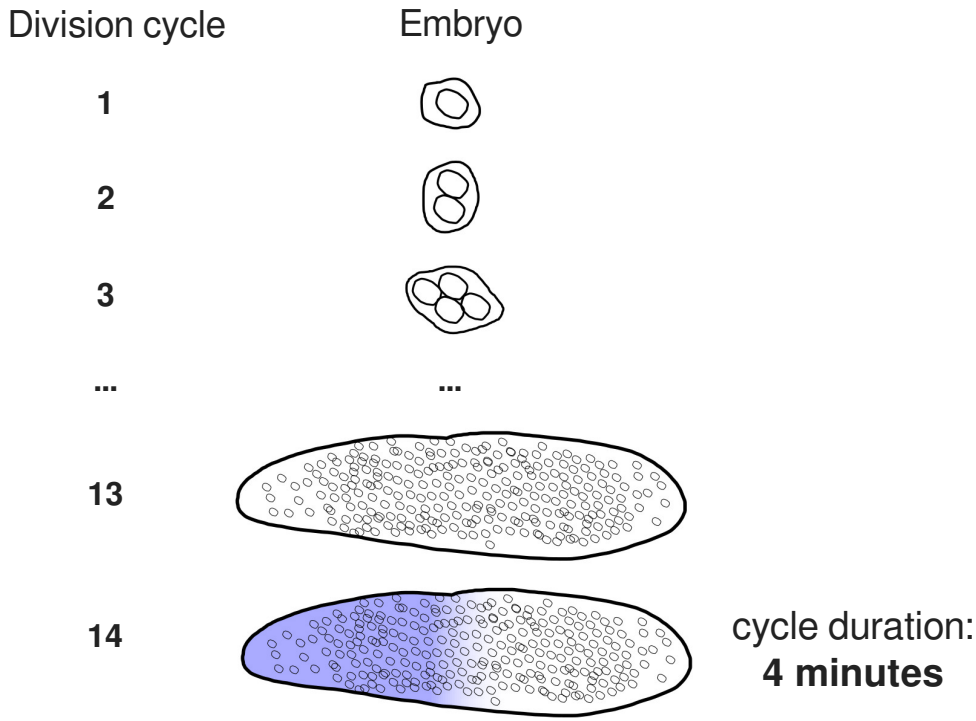
scene7

Precise sensing on a tight schedule



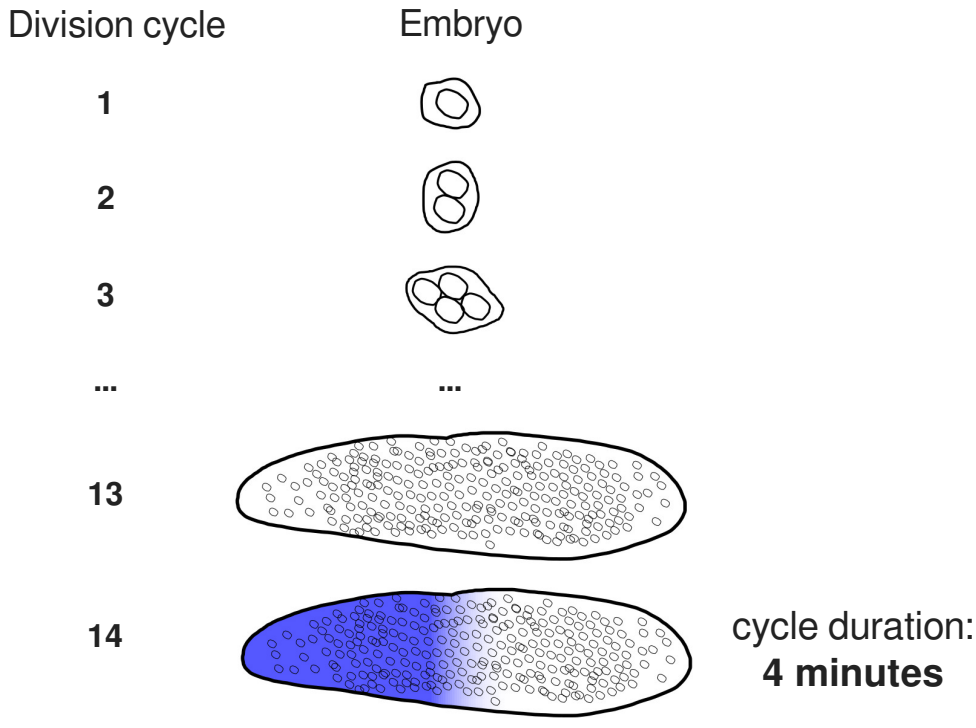
scene8_1

Precise sensing on a tight schedule



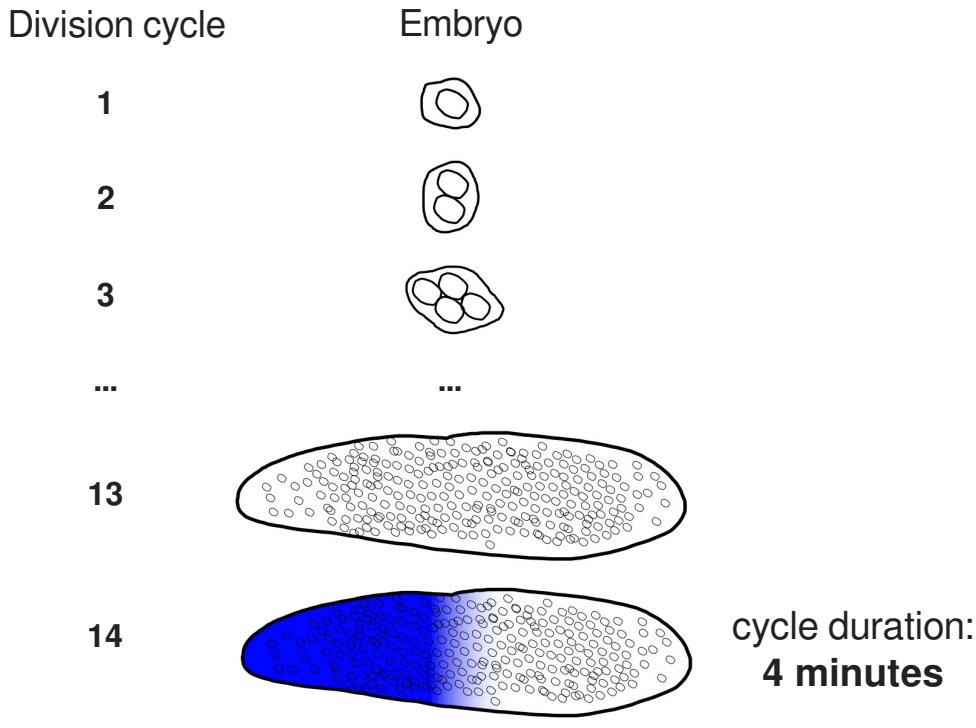
scene8$_2$

Precise sensing on a tight schedule



scene8_3

Precise sensing on a tight schedule



scene8_4

Understanding the regulatory precision of promoters

Goals
How to quantify precision, to evaluate/compare promoter models?
Which promoter architectures are optimal?
How are these optimal architectures shaped by their constraints and by the objective function they maximize?

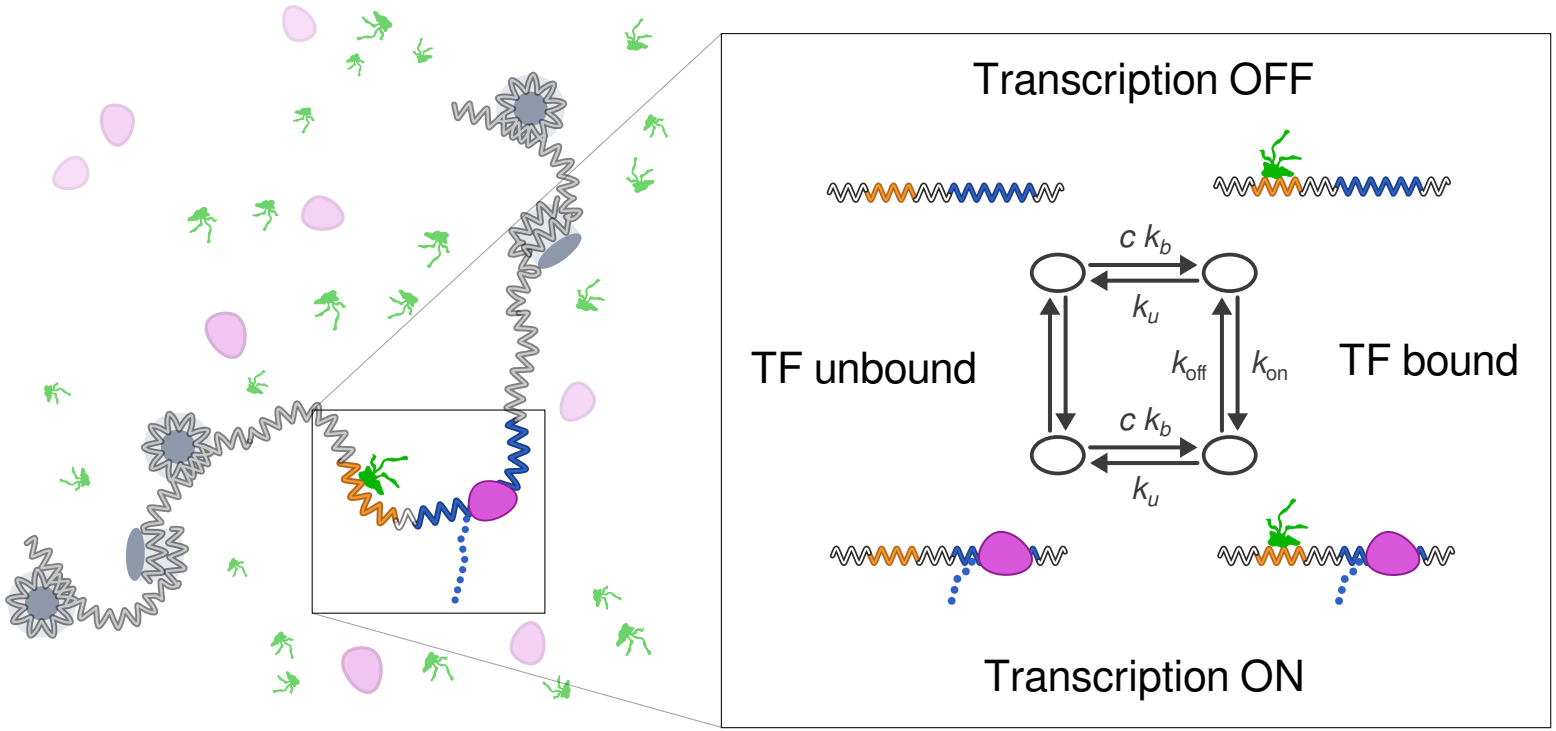
scene9

Understanding the regulatory precision of promoters

Approach
Defining a measure of precision
Optimizing precision under constraints

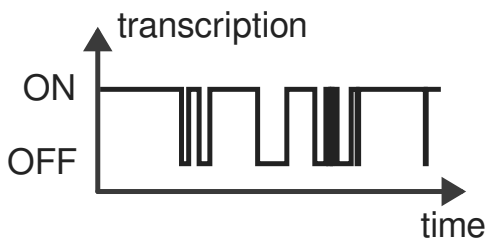
scene10

A simple model of transcriptional control



DNA
Promoter
Gene
Transcription factor
RNA Polymerase
mRNA

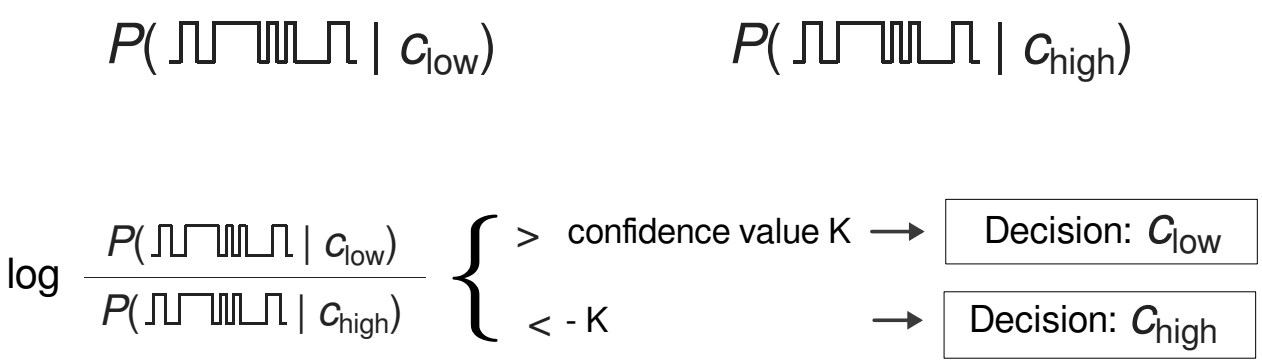
Concentration: high



low



Sequential hypothesis testing

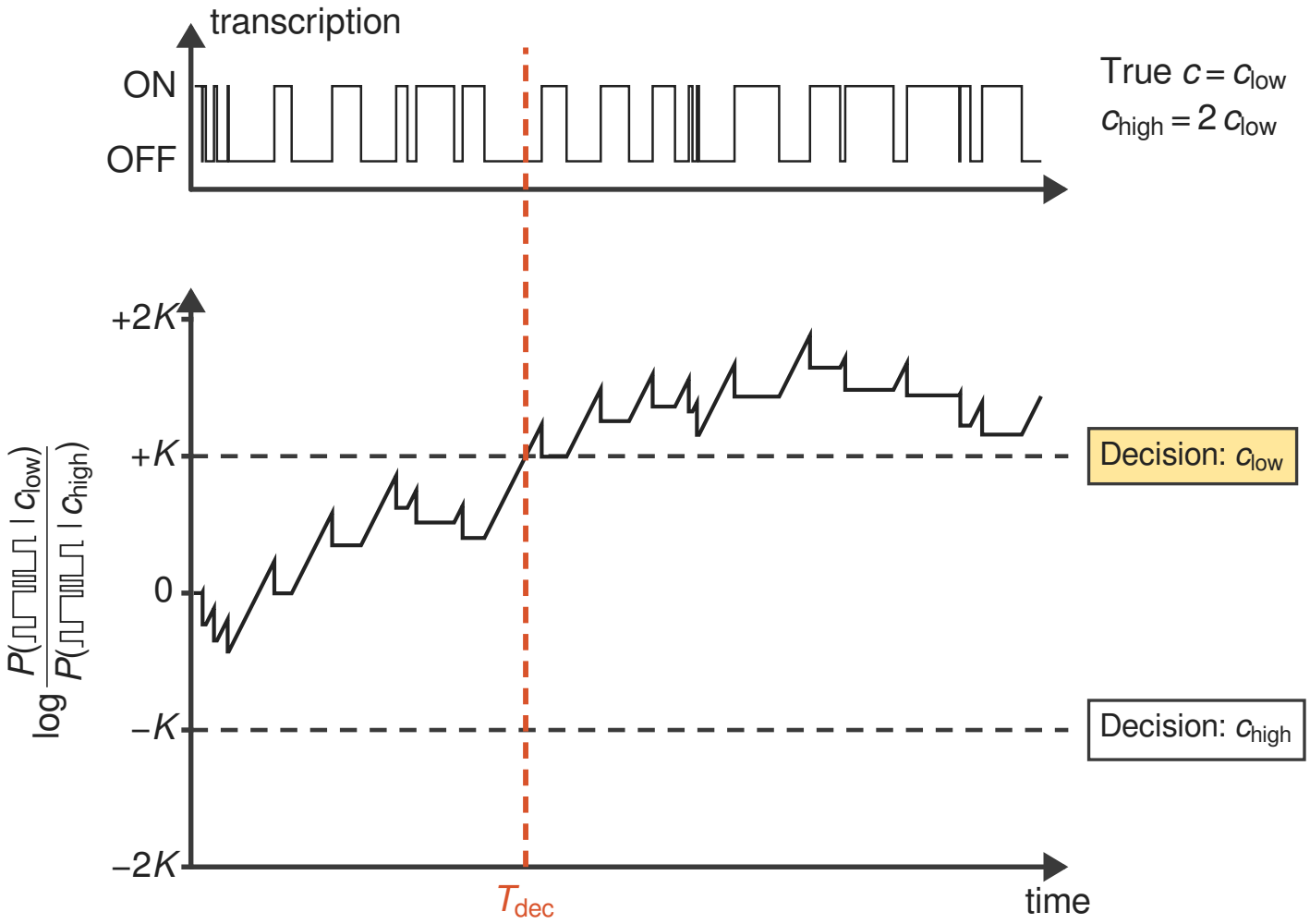


Sequential Probability Ratio Test (SPRT)

Literature:

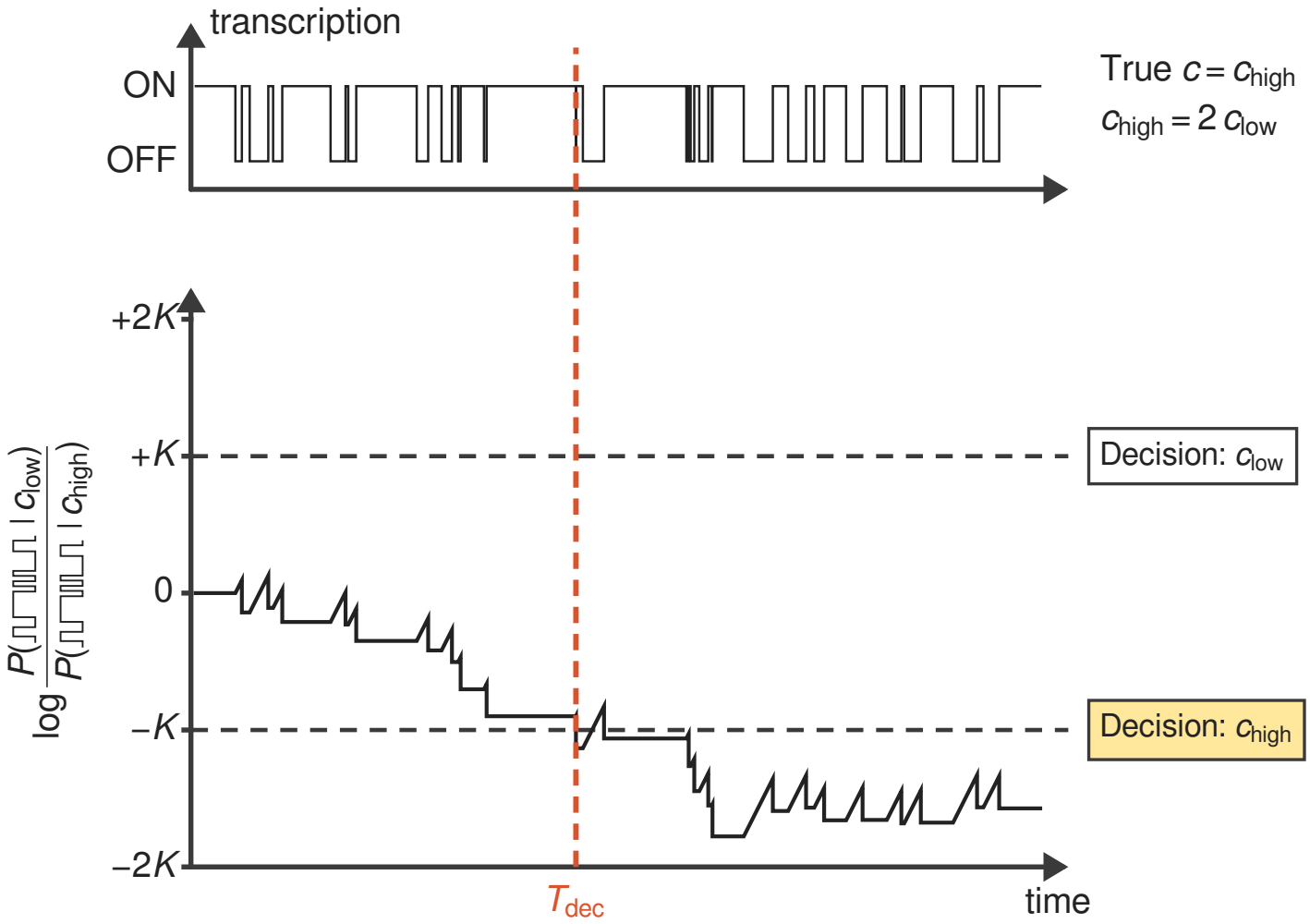
Wald 1949 (conception)
Siggia, Vergassola 2012 (application to biological receptors)
Desponds, Vergassola, Walczak 2022 (optimization)
Lammers, Flamholz, Garcia, 2024 (nonequilibrium receptors)

Sequential hypothesis testing



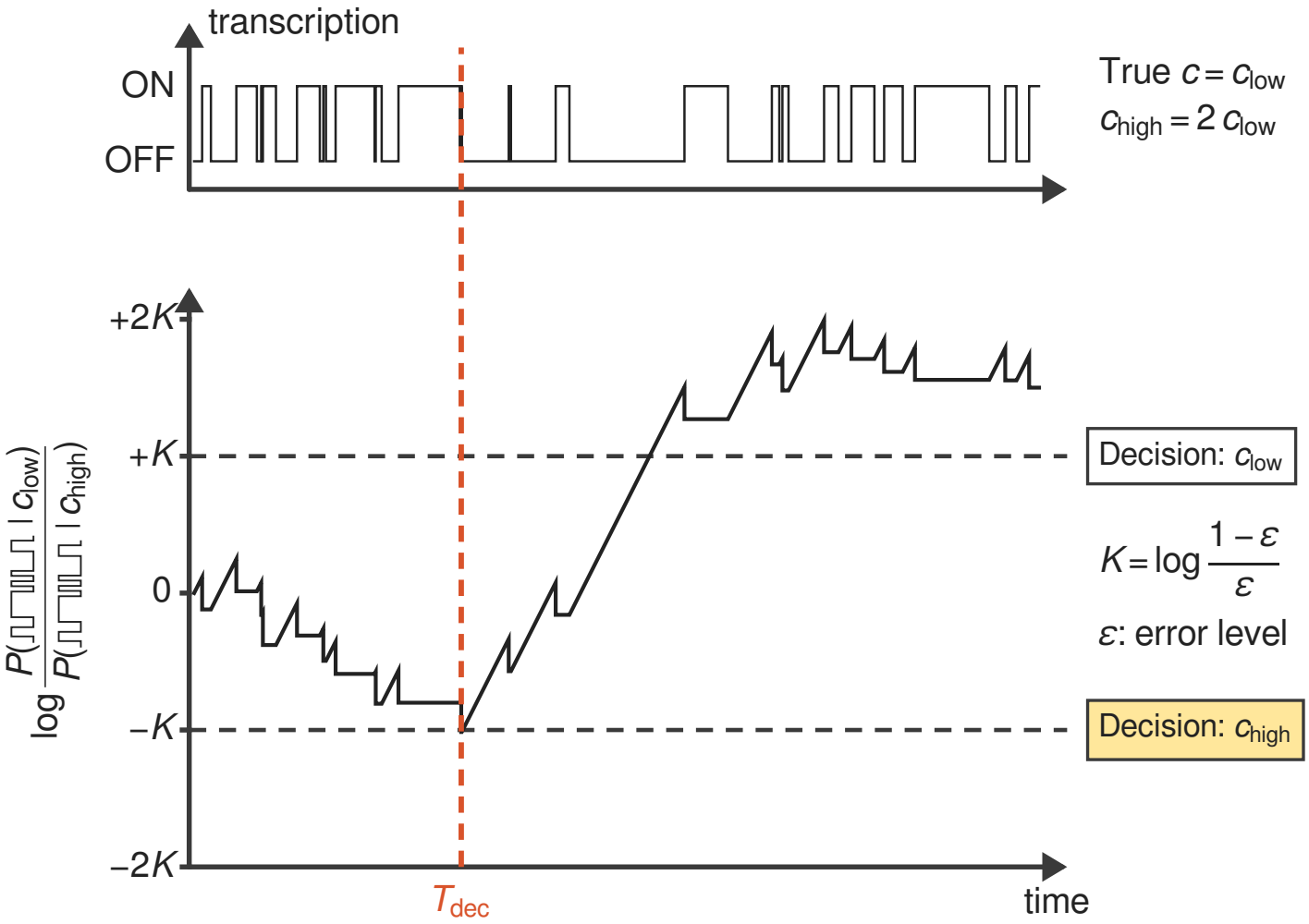
scene13_1

Sequential hypothesis testing



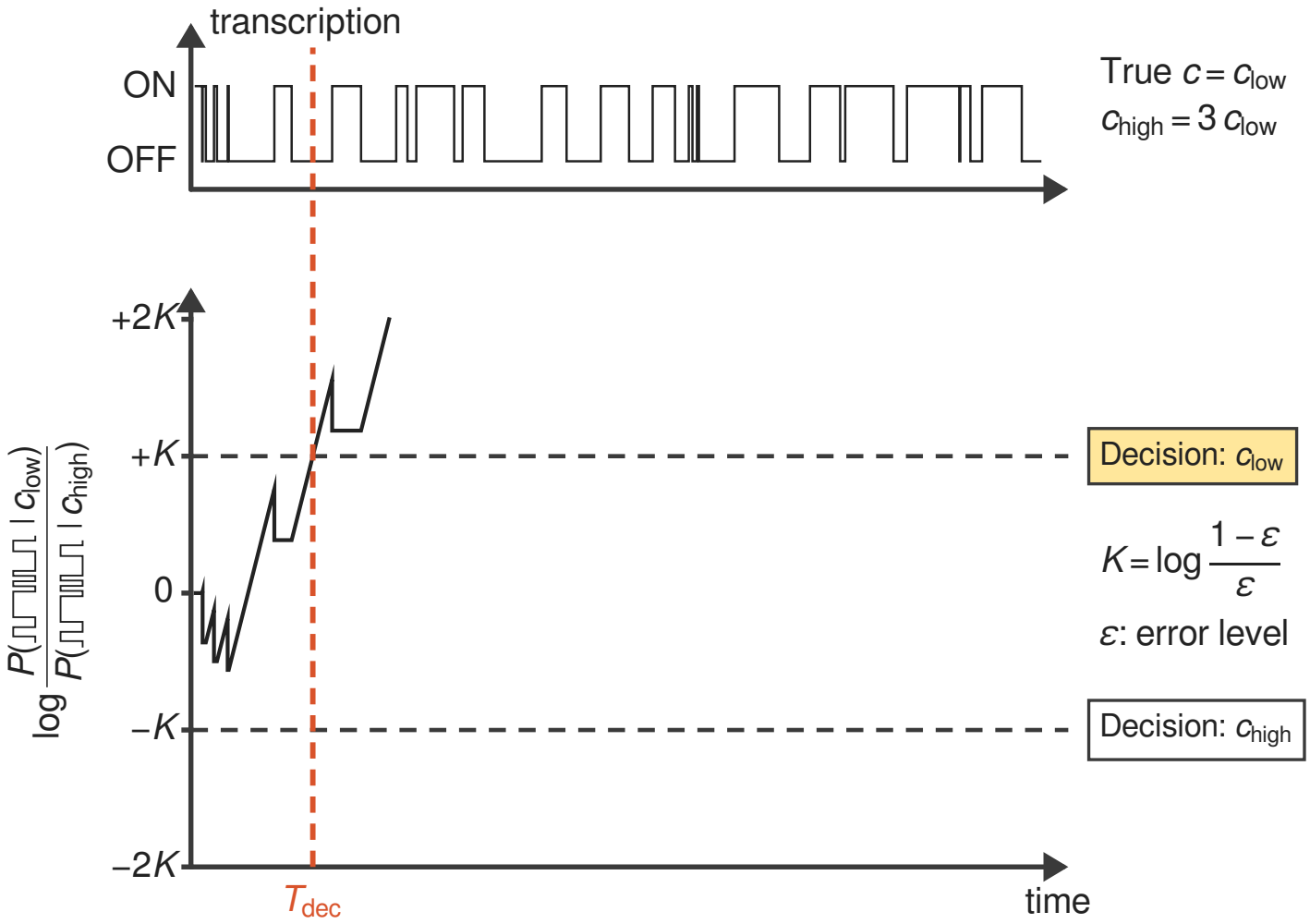
scene13_2

Sequential hypothesis testing



scene13_3

Sequential hypothesis testing



scene13$_4$

Fisher information rate and decision time

Drift-diffusion approximation when $\delta c/c \ll 1$

$$\frac{d\mathcal{R}}{dt} = \underbrace{V}_{\text{Drift}} + \underbrace{\sqrt{2V}\eta}_{\text{Diffusion}} \leftarrow \text{Gaussian noise } \langle \eta(t)\eta(t') \rangle \sim \delta(t-t')$$

SPRT mean decision time (Siggia & Vergassola 2012)

$$\langle T_{\text{dec}} \rangle = \frac{K}{V} \tanh\left(\frac{K}{2}\right)$$

$$V \approx \frac{1}{2}\left(\frac{\delta c}{c}\right)^2 \dot{\mathcal{F}}(c)$$

$$\dot{\mathcal{F}}(c) = -\lim_{t\to\infty} \frac{1}{t} \left\langle \frac{\partial^2 \mathcal{L}(c)}{(\partial \ln c)^2} \right\rangle_c$$

$$\mathcal{L}(c;t) = \ln P(O(t)|c)$$

scene14$_1$

The optimization problem

$$\text{Maximize } \dot{\mathcal{F}}(c)$$

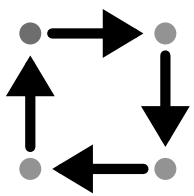
subject to the constraints

$$k_b \leq 4aD \text{ and } k_{\text{u,on,off}} \leq k_{\max}$$

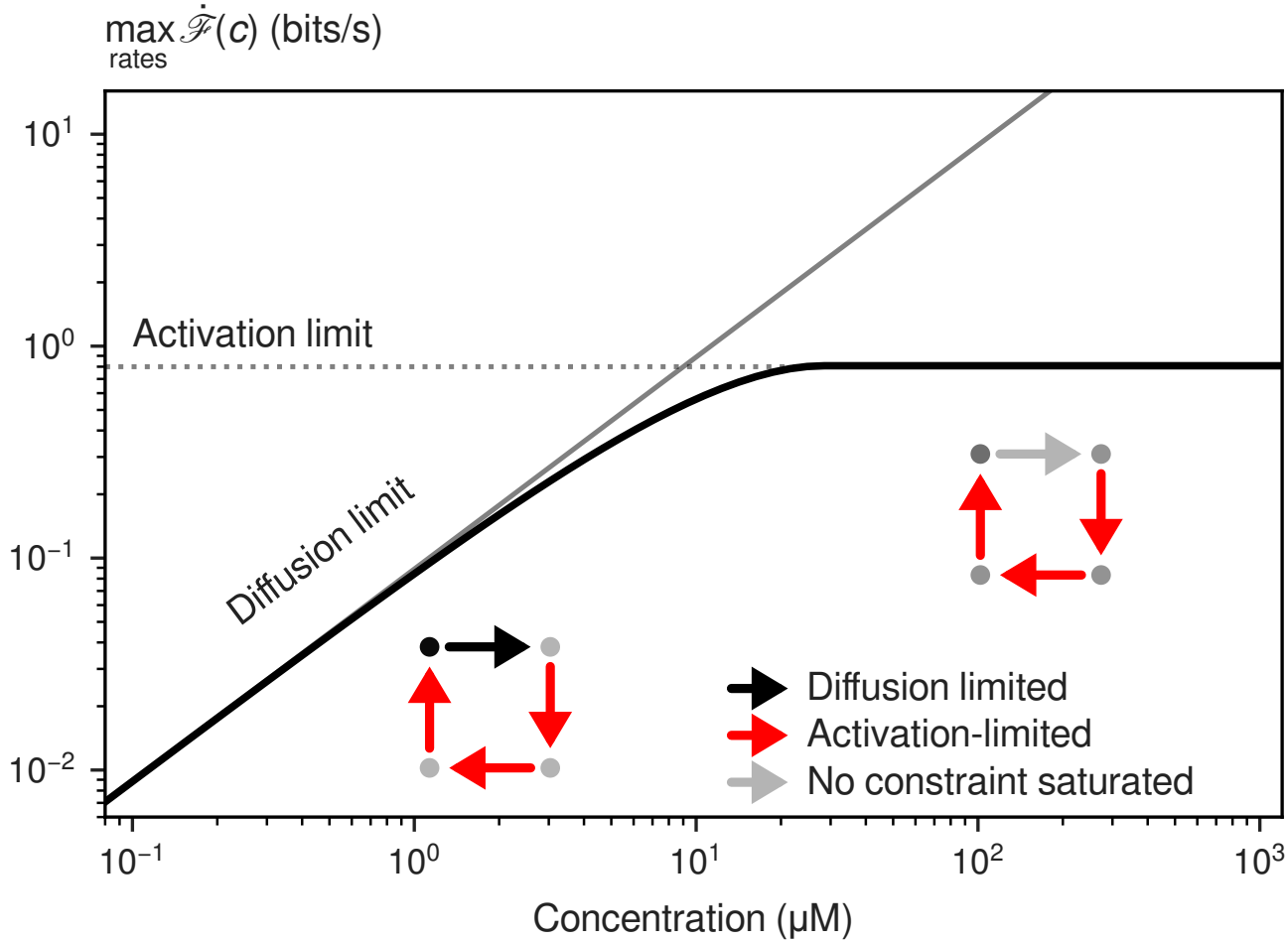
Diffusion Activation

In practice:

1. numerical first pass (dynamic programming)
2. ansatz obtained from solutions (analytical)

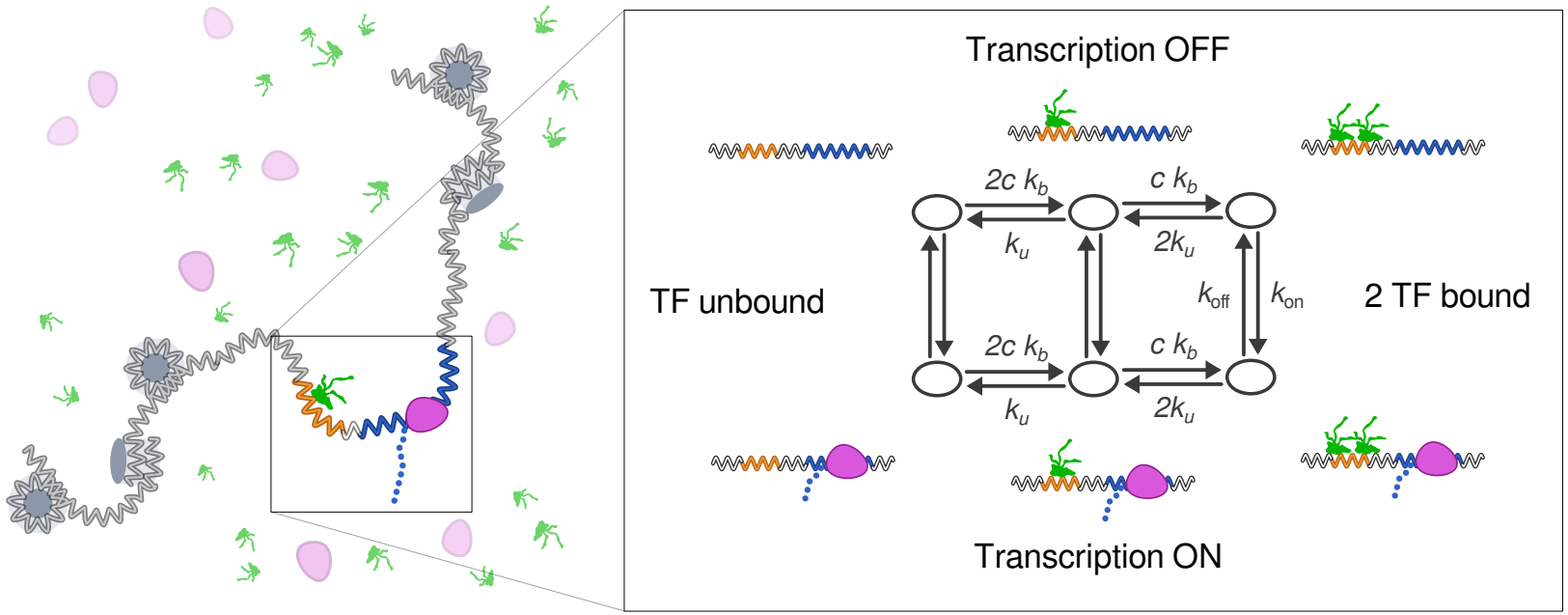


Optimal promoters: two distinct regimes



scene16

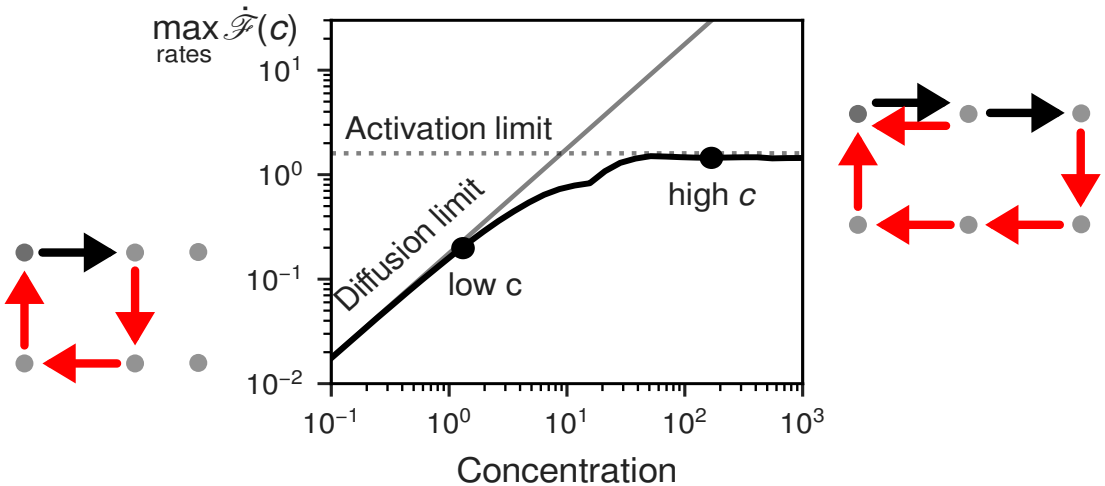
Cooperative binding



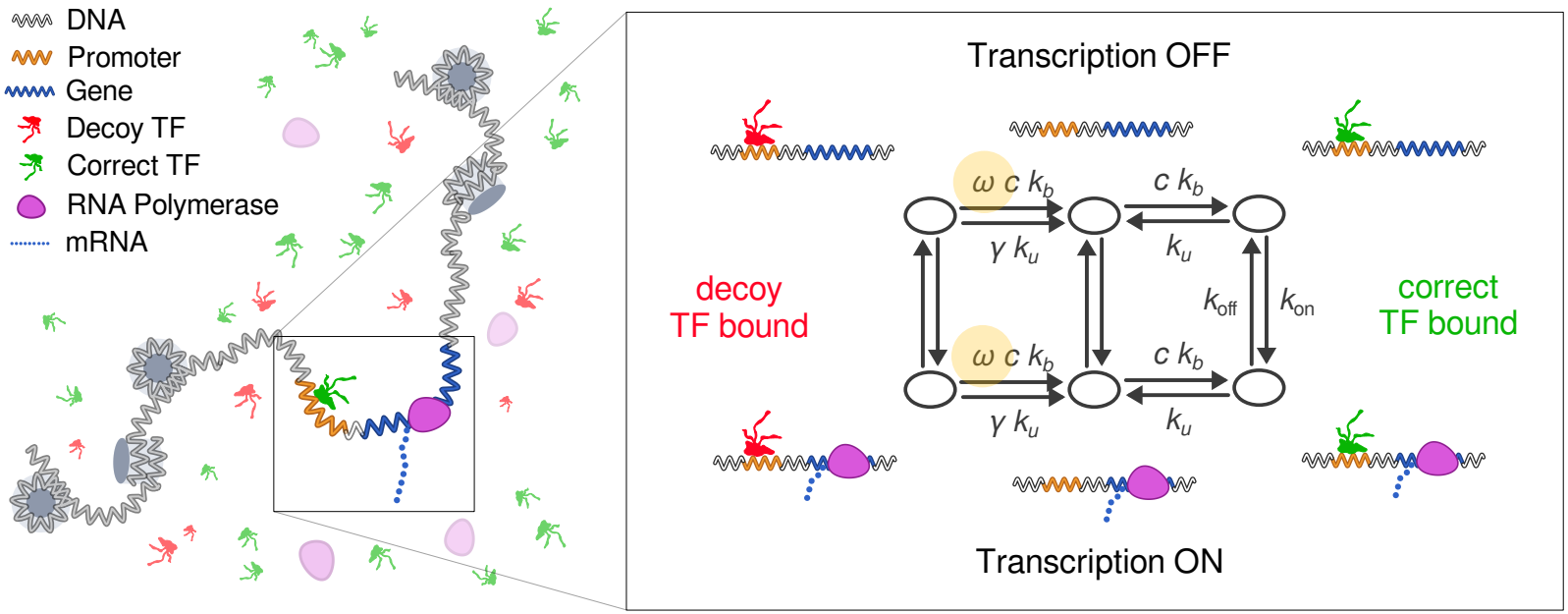
DNA
Promoter
Gene
Transcription factor
RNA Polymerase
mRNA

Diffusion limited
Activation-limited
No constraint saturated

↓ Maximize $\dot{\mathcal{F}}(c)$ subject to the constraints ↓



Spurious binding



$\downarrow$ Maximize $\dot{\mathcal{F}}(c)$ subject to the constraints $\downarrow$

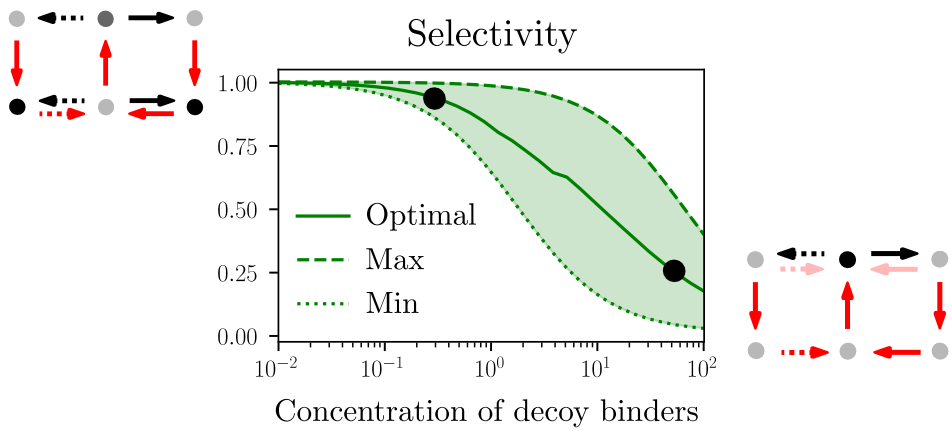
ω : [decoy TF] / [correct TF]

$$\gamma : \frac{\text{decoy unbinding rate}}{\text{correct unbinding rate}}$$

Selectivity f = P(transcribing | correct TF)

$$\max f = \gamma^2 / (\omega + \gamma^2)$$

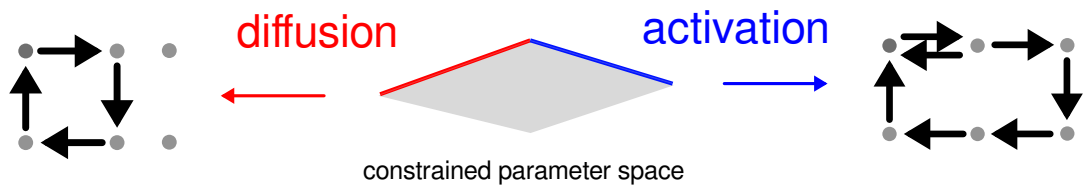
$$\min f = \gamma / (\omega(\gamma+1)/2 + \gamma)$$



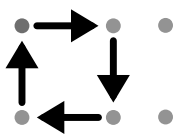
scene18

Takeaways

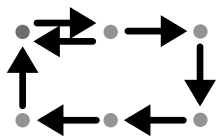
The limiting constraint shapes the optimal architecture



Solution set is disconnected: each cluster points to a different sensing strategy



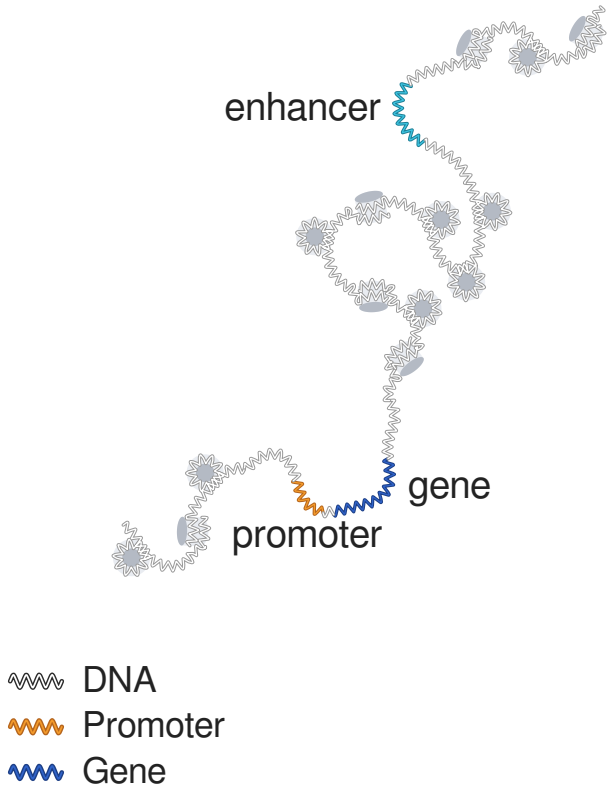
count rare
binding events



leverage
cooperative binding

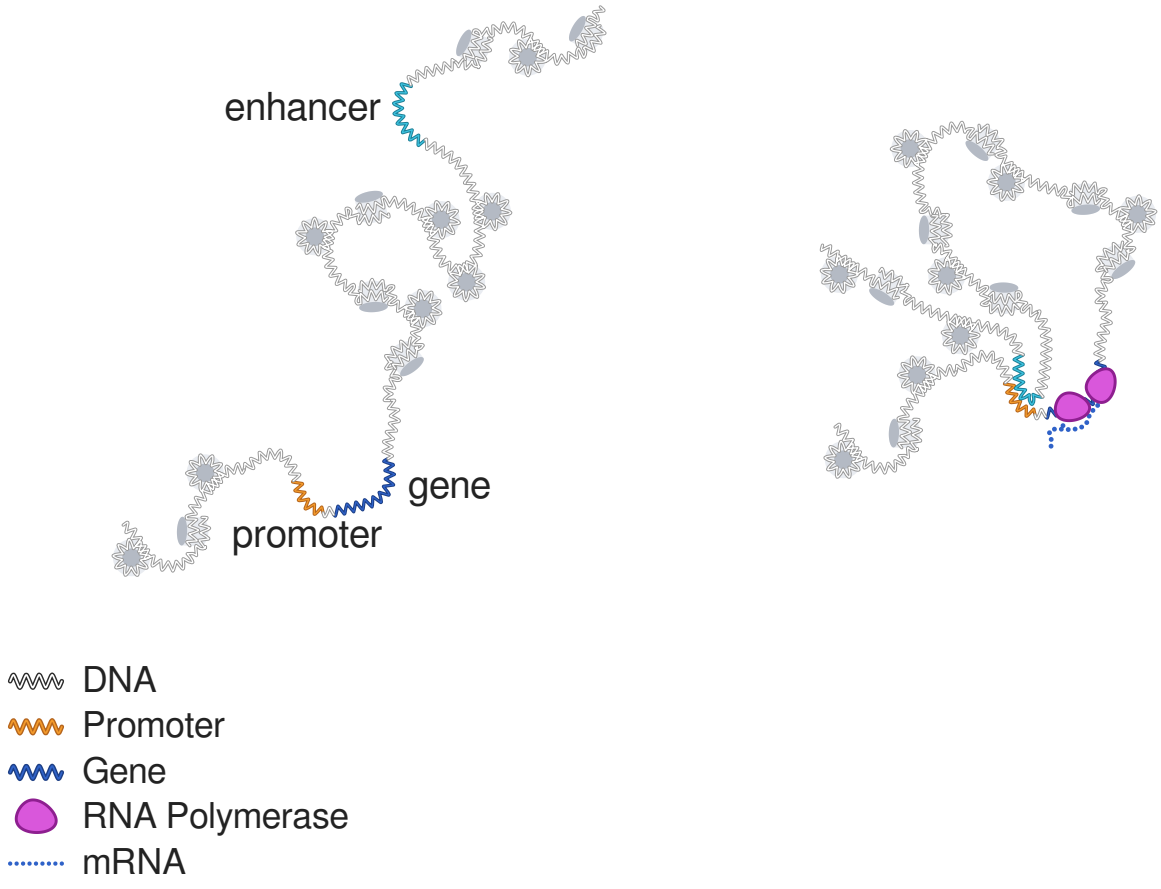
Sharpness, selectivity do not lie at extrema

Transcriptional control by distal elements



scene20_1

Transcriptional control by distal elements

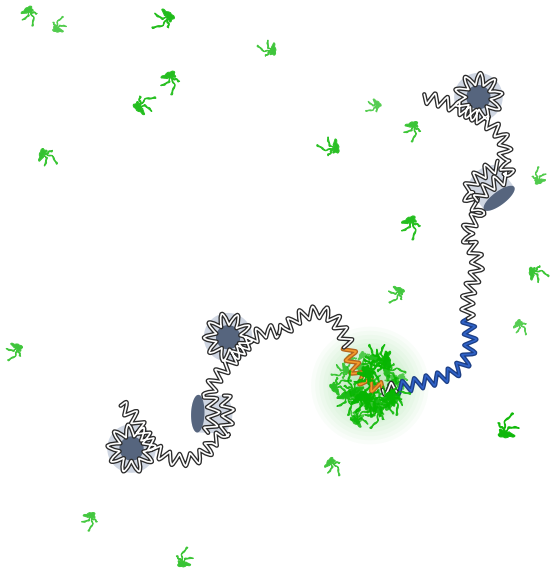


scene20_2

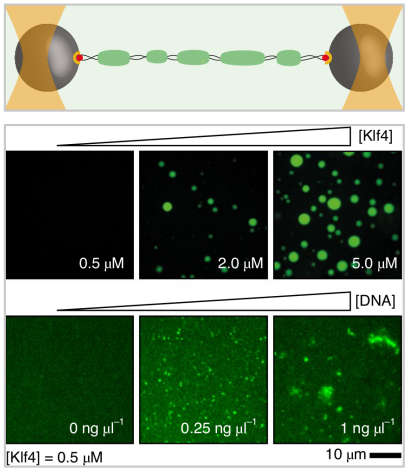
Transcriptional control through liquid phase separation?

scene21$_1$

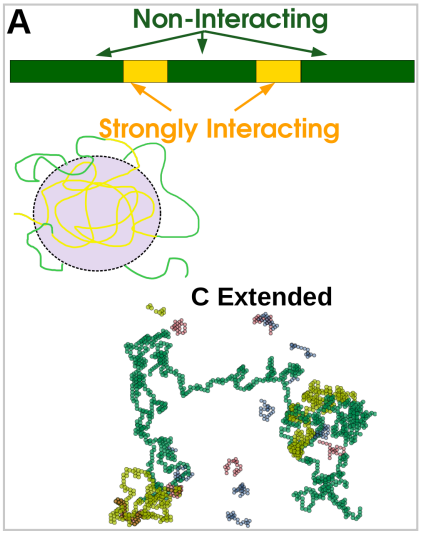
Proteins condense on the surface of DNA



DNA
Promoter
Gene
Transcription factor

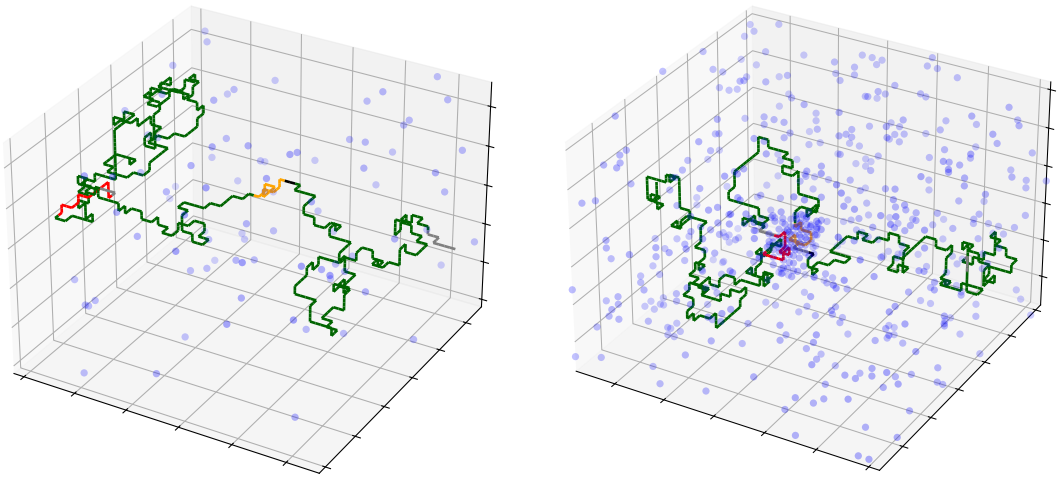


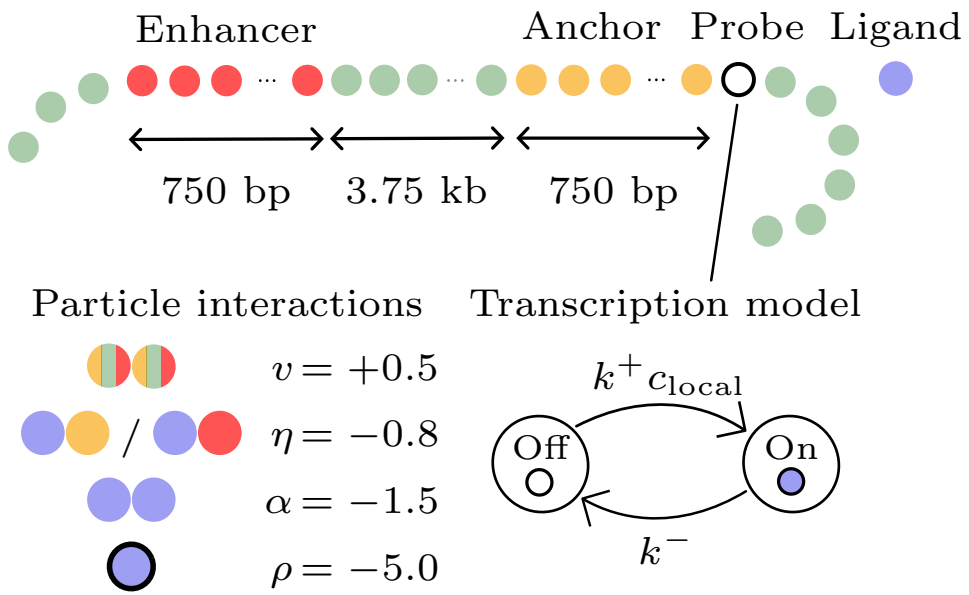
Sequence-dependent surface condensation of a pioneer transcription factor on DNA
Morin et al., 2022



Protein-DNA Co-condensation is Prewetting to a Collapsed Polymer
Rouches & Machta, 2025

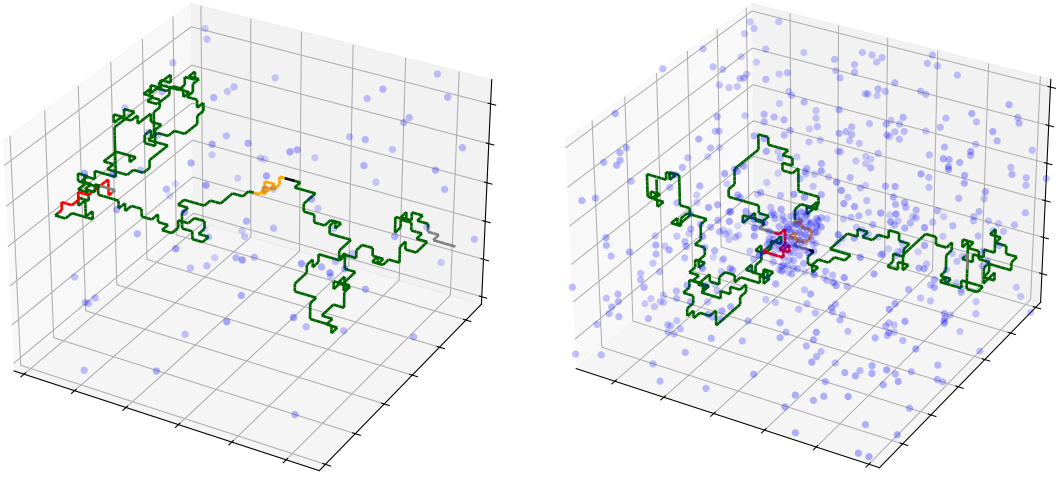
Coupling condensation and polymer dynamics

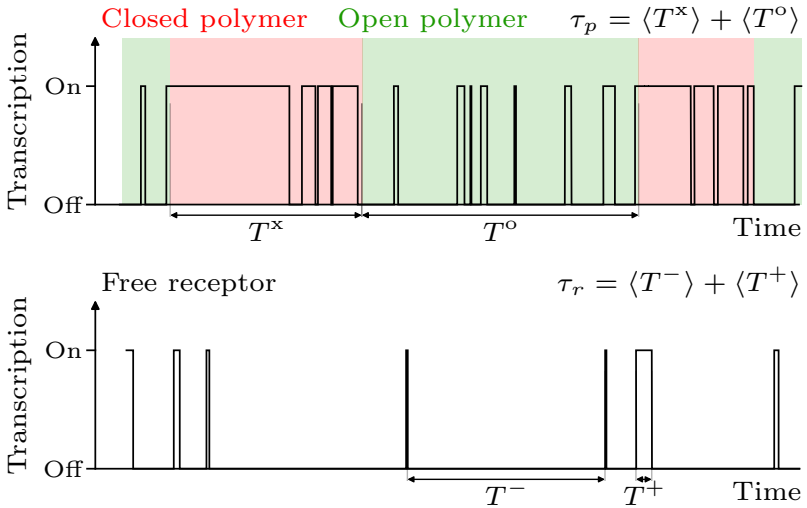


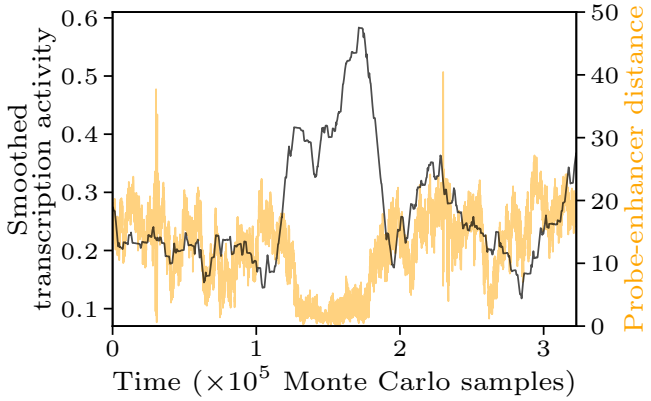


scene22

Coupling condensation and polymer dynamics

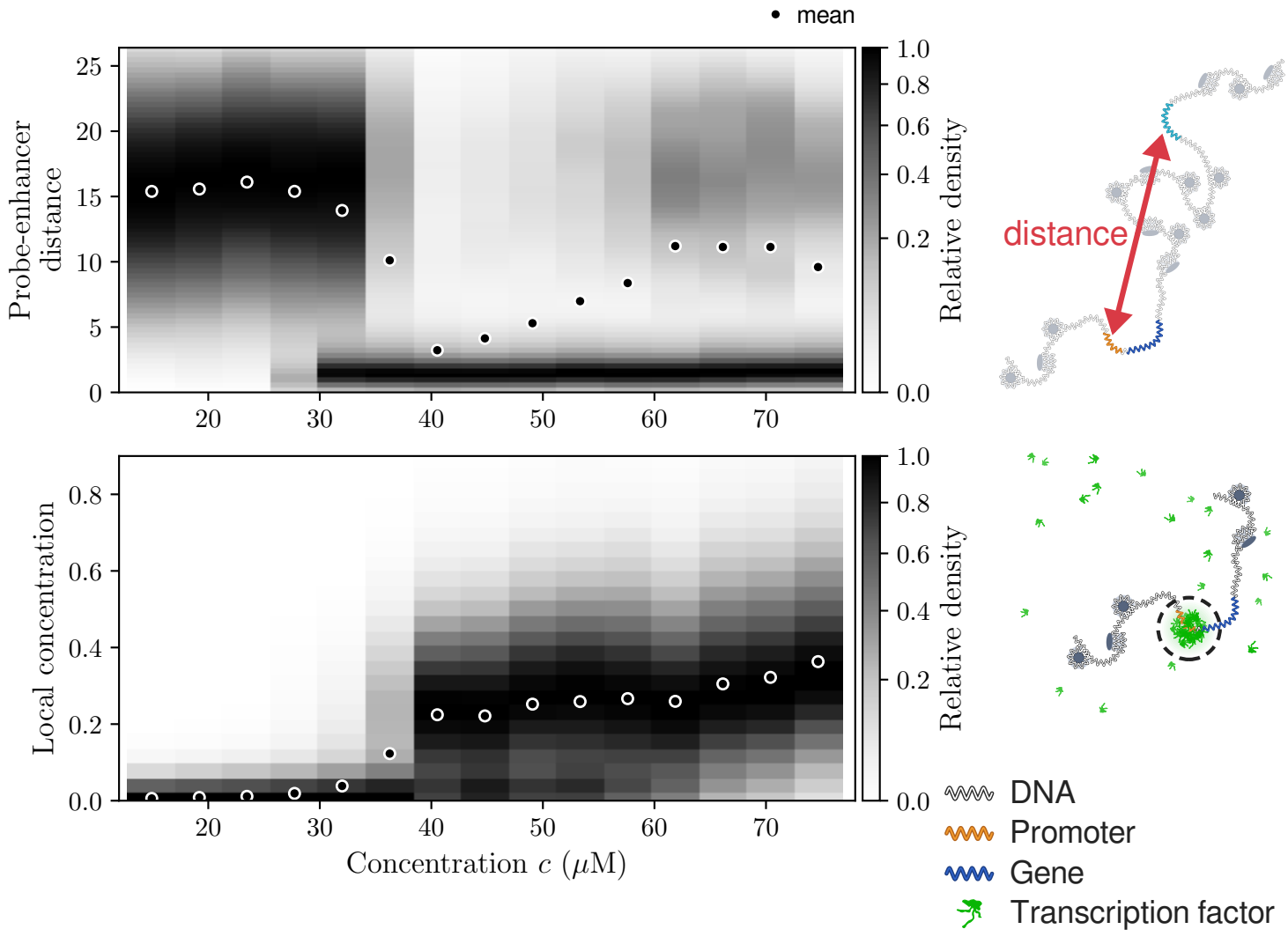






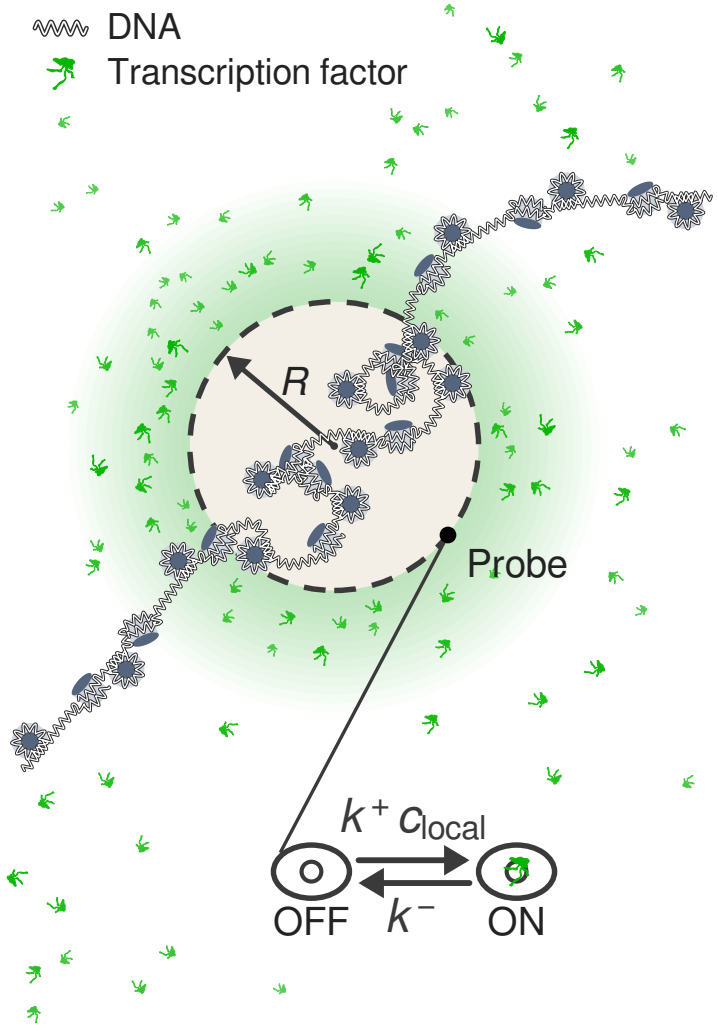
scene23$_2$

Polymer configuration depends sensitively on concentration



scene24

Continuous model of surface condensation



Flory–Huggins potential

$$f(\phi) = \frac{\phi}{v}\ln\phi + (1-\phi)\ln(1-\phi) - \chi\phi^2 - \mu\phi$$

Cahn–Hilliard equation

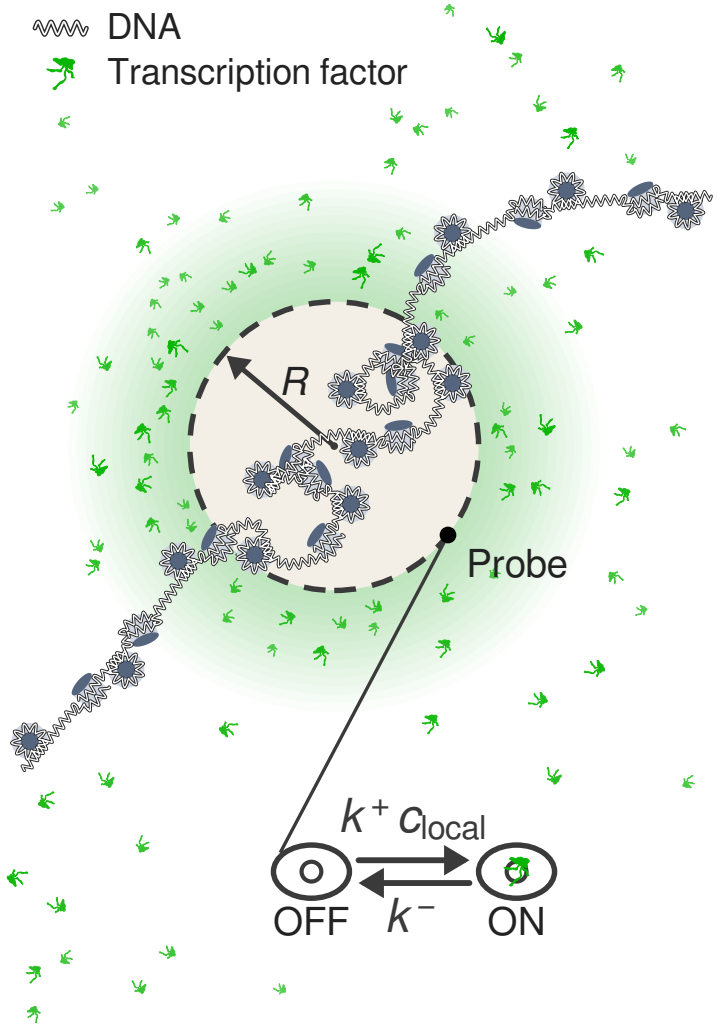
$$0 = \nabla^2\left[f'(\phi) - \kappa\nabla^2\phi\right]$$

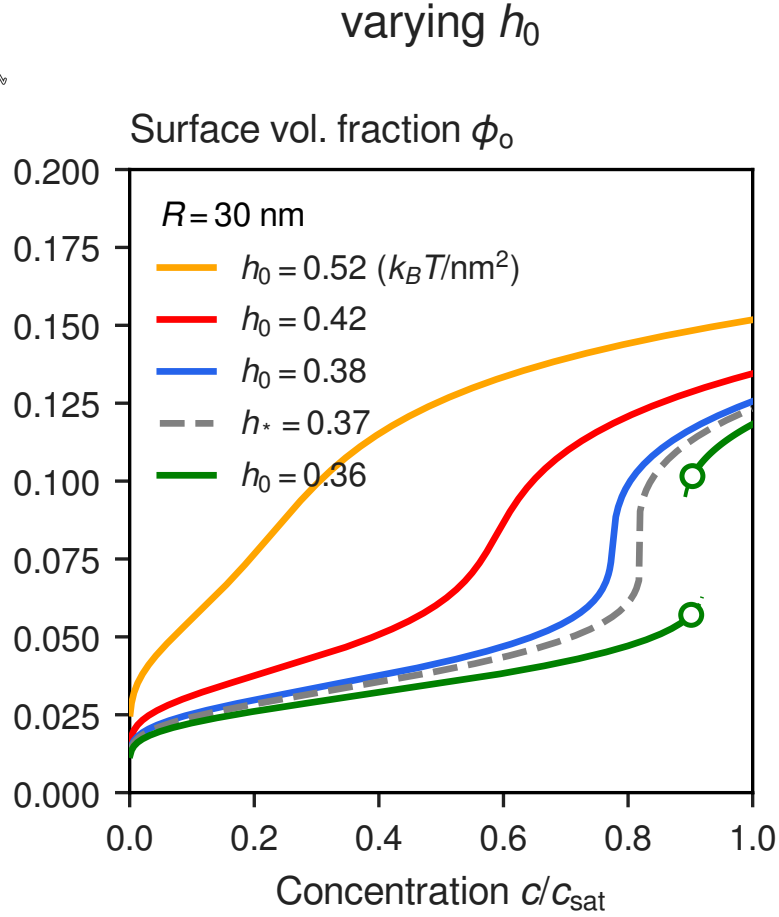
$$\kappa\,\partial_r\phi\,|_{r=R} = -h_0$$

ϕ	TF volume fraction
v	TF / solvent volume ratio
χ	TF–TF attraction
μ	chemical potential
h_0	surface attraction
κ	interface stiffness

scene25$_1$

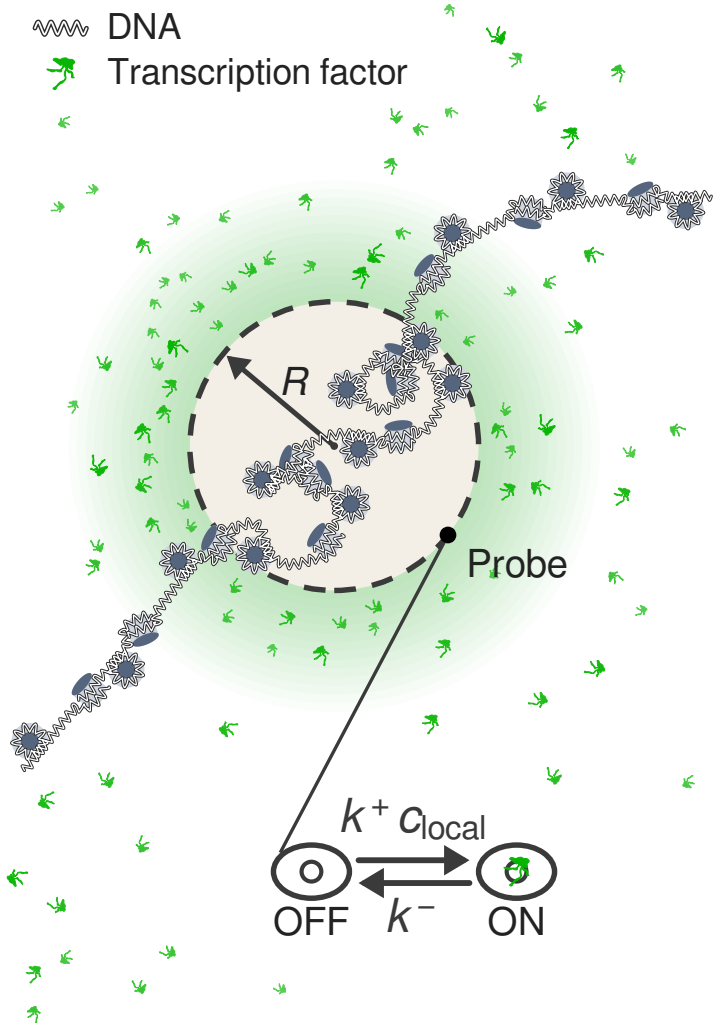
The prewetting transition: varying surface attraction

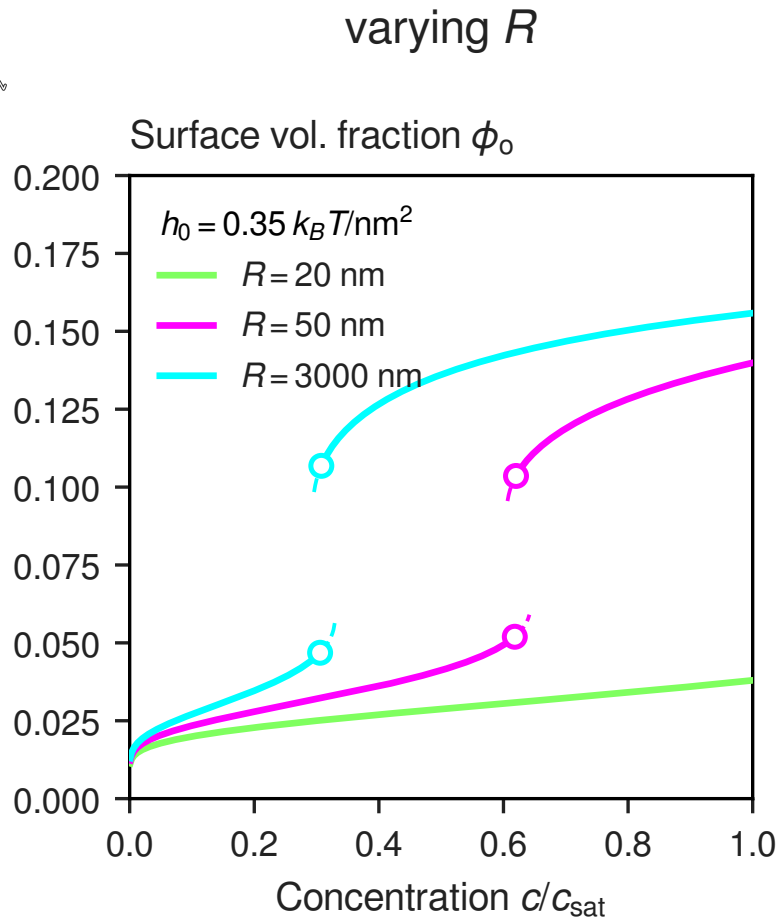




scene25$_2$

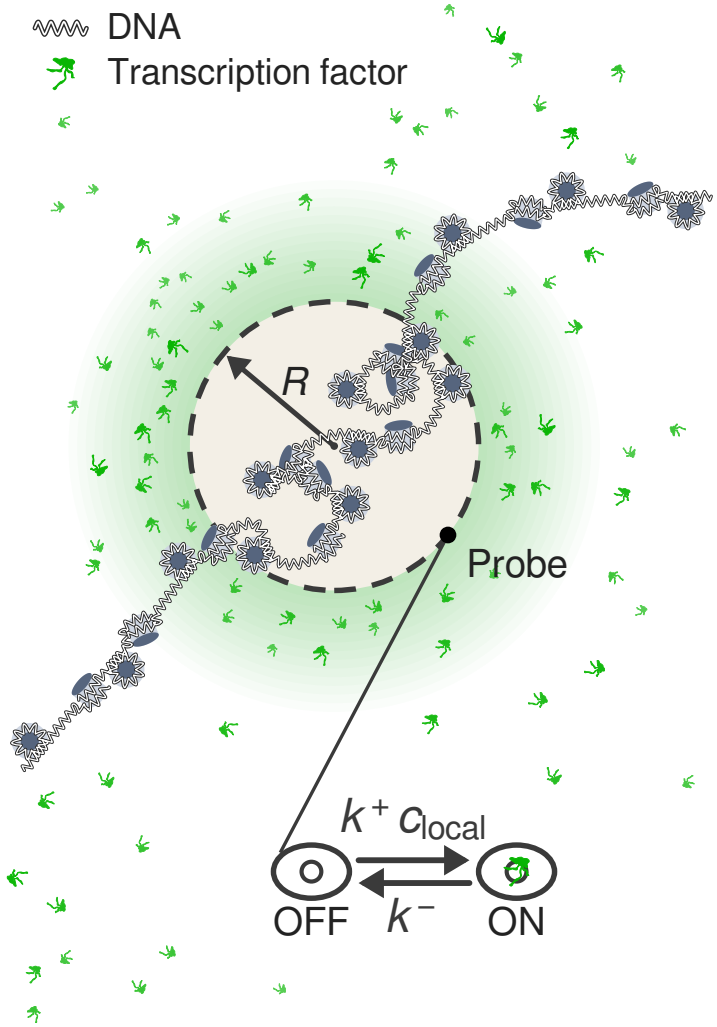
The prewetting transition: varying surface curvature





scene25$_3$

Quantifying sensitivity of local concentration



Free probe

$$\dot{\mathscr{F}}_{free}(c) = \frac{1}{\tau_r(c)}$$

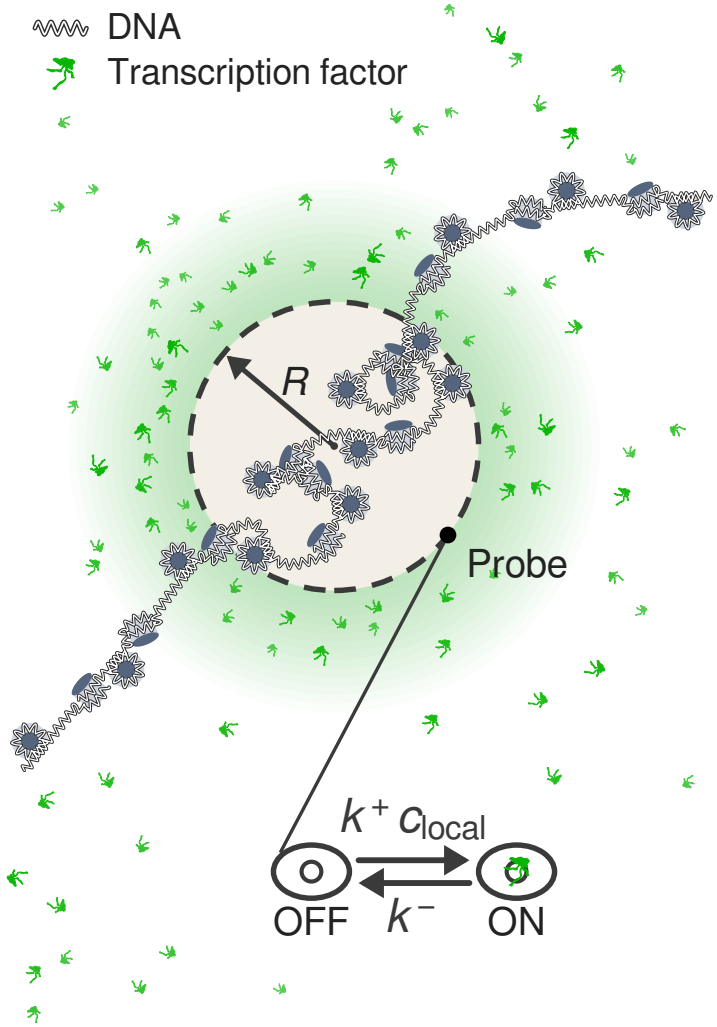
Probe on the anchor

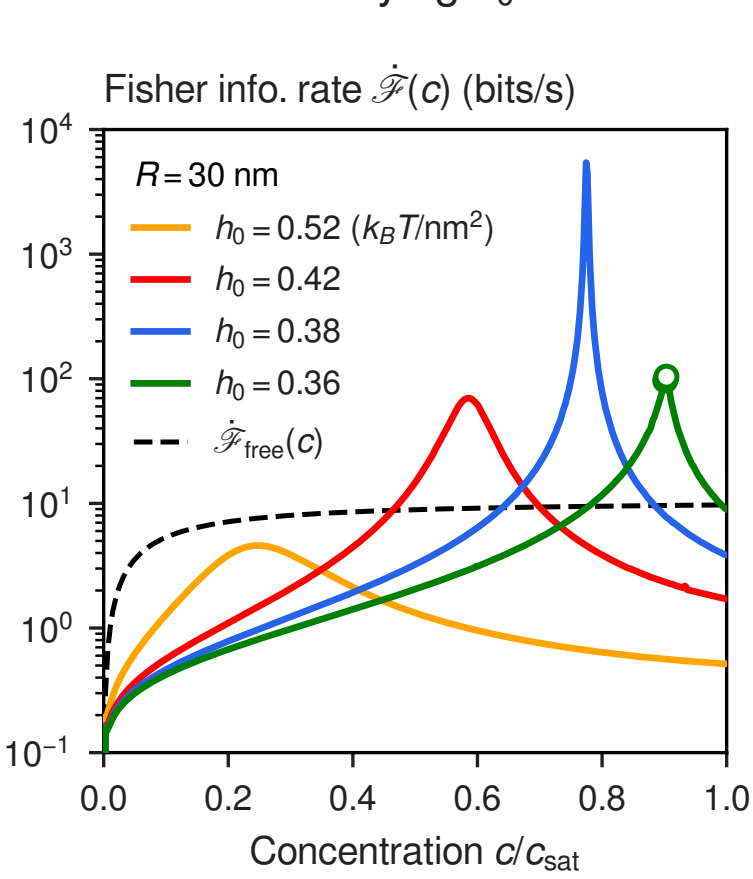
$$\dot{\mathscr{F}}(c) = \underbrace{\frac{1}{\tau_r(\phi)}}_{\text{probe}} \underbrace{\left(\frac{d\log\phi}{d\log c}\right)^2}_{\text{surface}}$$

τ_r mean cycle time of the probe
ϕ TF volume fraction at the probe

scene26$_1$

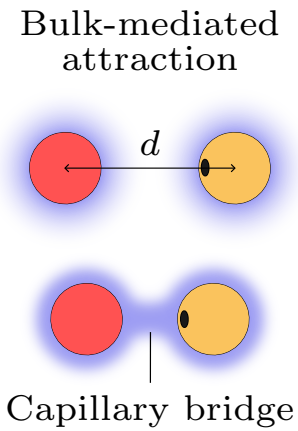
Quantifying sensitivity of local concentration

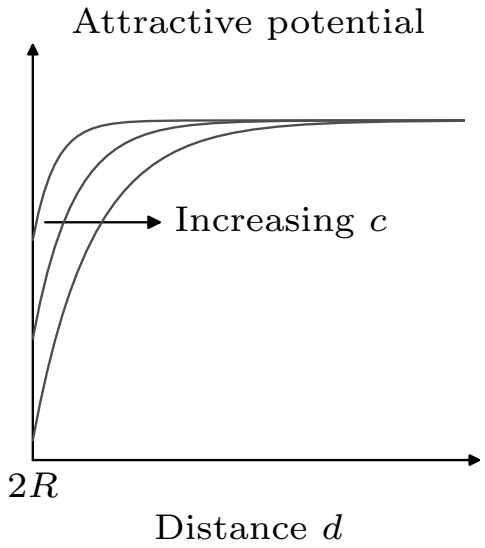


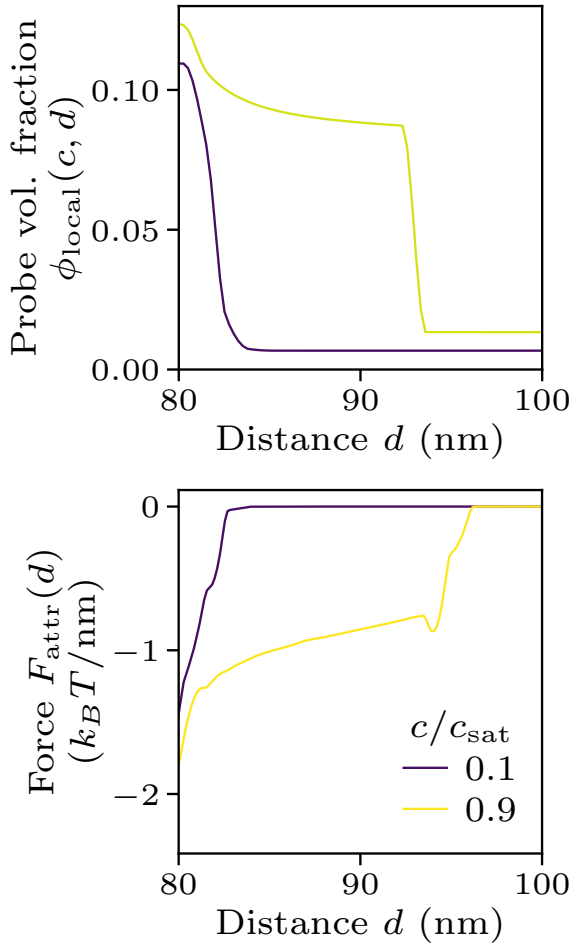


scene26_2

Bulk-mediated attraction

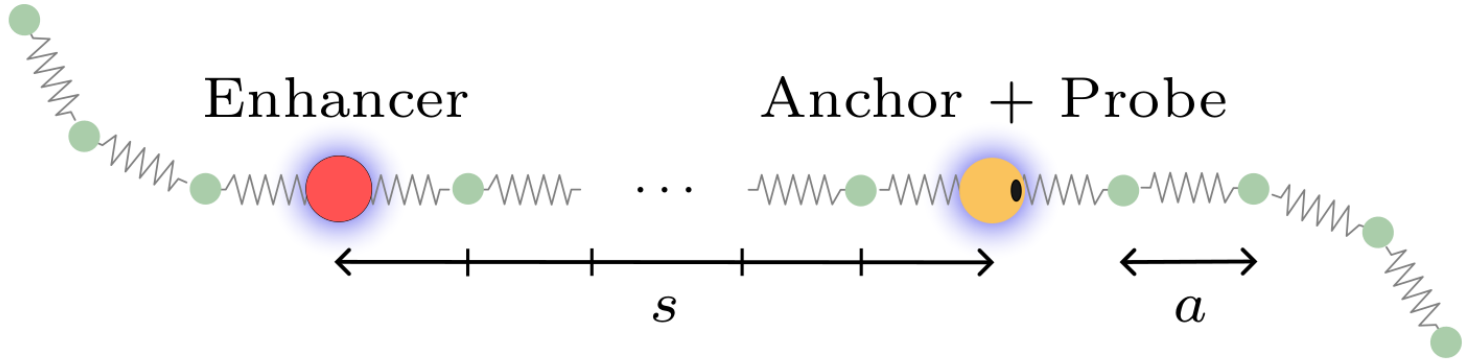






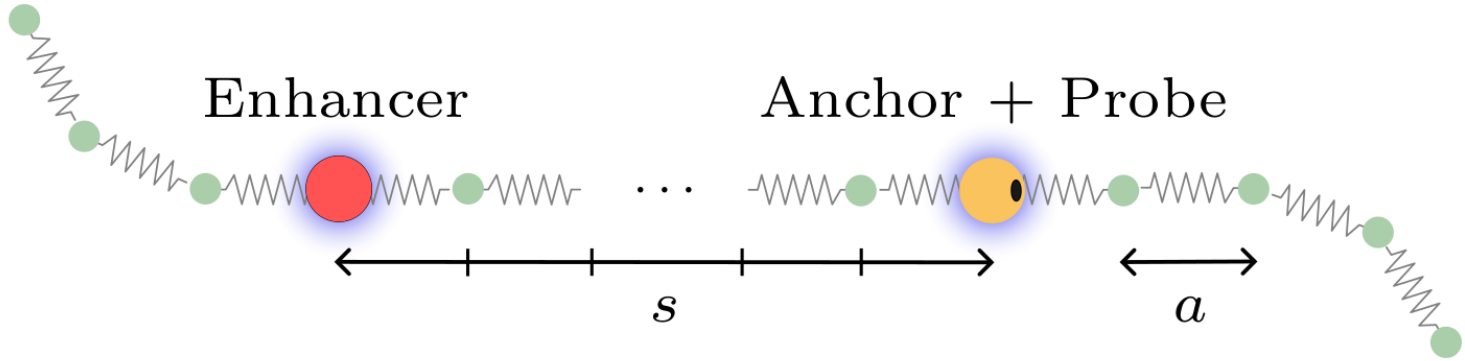
scene27

Modeling the polymer



$$P_{\text{poly}}(d) \propto d^2 e^{-3d^2/2sa^2} \propto e^{-U_{\text{poly}}(d)/k_B T} \tag{1}$$

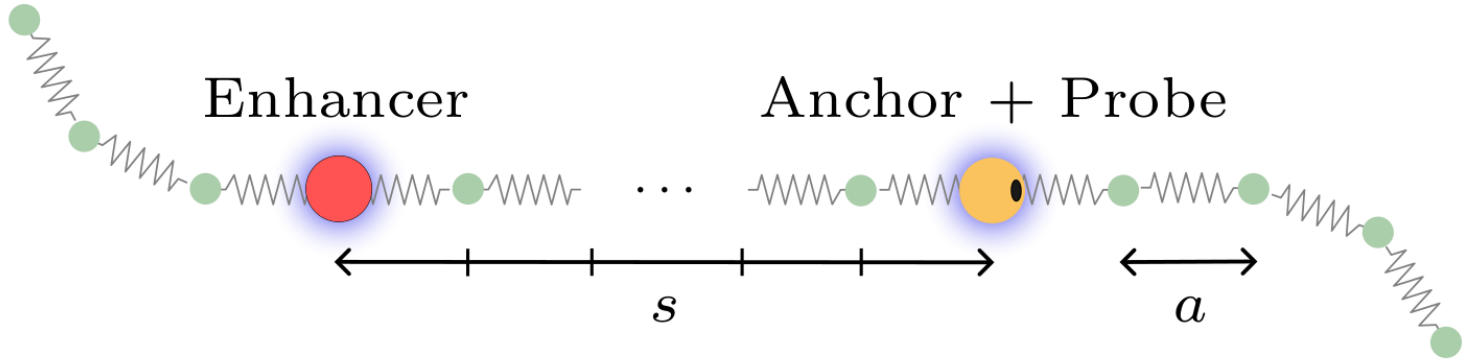
Modeling the polymer



$$P_{\text{poly}}(d) \propto d^2 e^{-3d^2/2sa^2} \propto e^{-U_{\text{poly}}(d)/k_B T} \tag{1}$$

$$U_{\text{tot}}(d) = U_{\text{attr}}(d) + U_{\text{poly}}(d) \tag{2}$$

Modeling the polymer

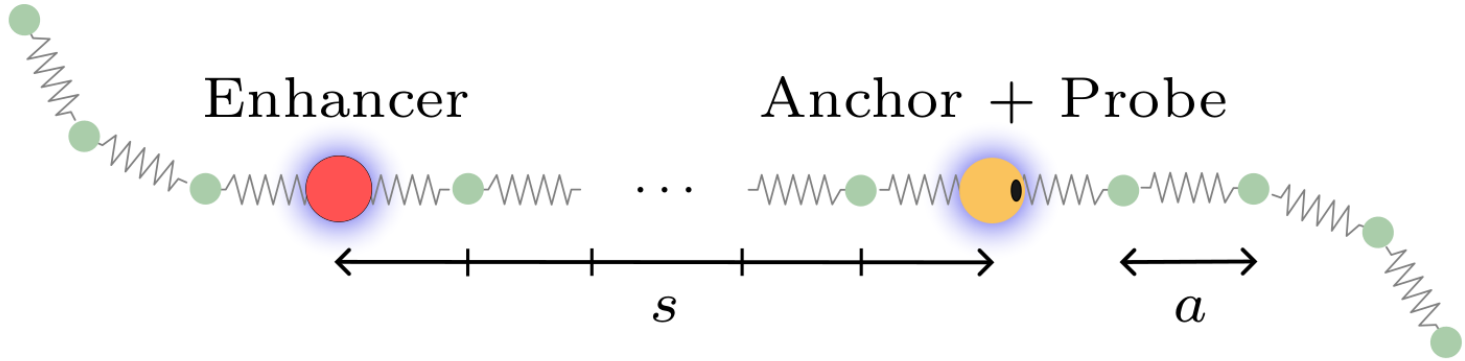


$$P_{\text{poly}}(d) \propto d^2 e^{-3d^2/2sa^2} \propto e^{-U_{\text{poly}}(d)/k_B T} \tag{1}$$

$$U_{\text{tot}}(d) = U_{\text{attr}}(d) + U_{\text{poly}}(d) \tag{2}$$

$$P_{\text{eq}}(d) \propto e^{-U_{\text{tot}}(d)/k_B T} \tag{3}$$

Modeling the polymer



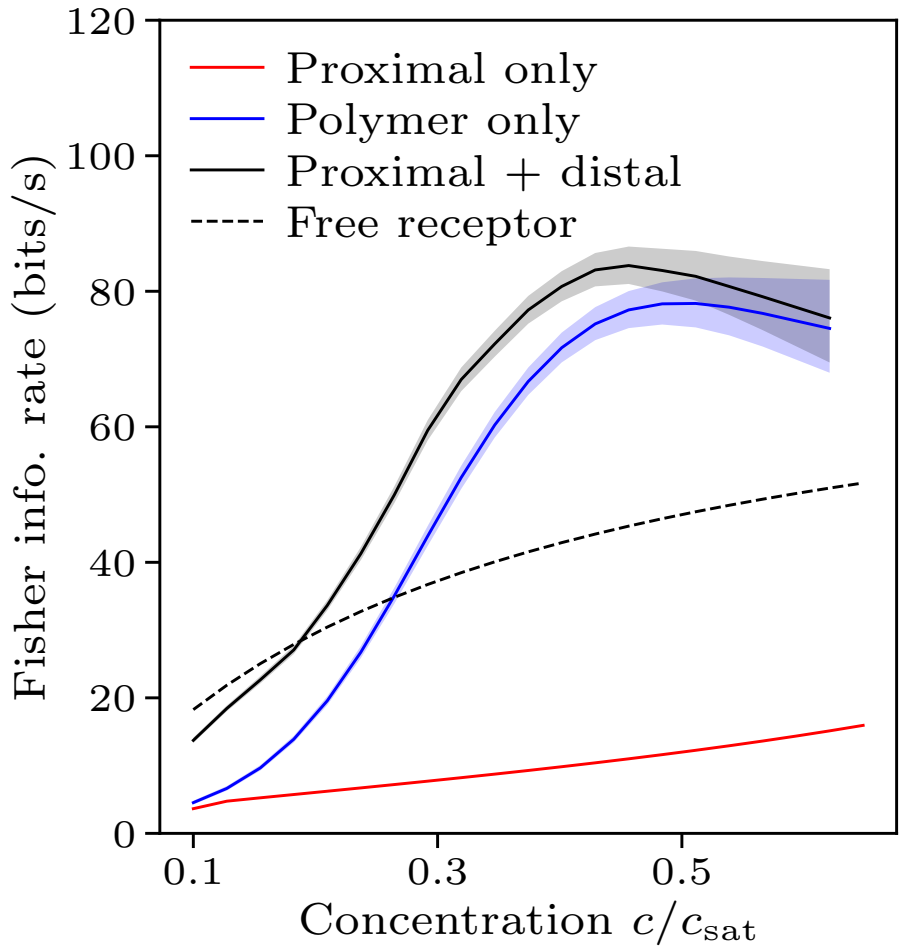
$$P_{\text{poly}}(d) \propto d^2 e^{-3d^2/2sa^2} \propto e^{-U_{\text{poly}}(d)/k_B T} \tag{1}$$

$$U_{\text{tot}}(d) = U_{\text{attr}}(d) + U_{\text{poly}}(d) \tag{2}$$

$$P_{\text{eq}}(d) \propto e^{-U_{\text{tot}}(d)/k_B T} \tag{3}$$

a: Kuhn bond length
s: genomic separation
d: euclidean distance

Sensitivity from polymer dynamics



scene30

Conclusions

- .